68. Video Input Module (VIN)

68.1 Overview

The video input module (hereinafter abbreviated as VIN) is a video capture module that stores in external memory YCbCr-422 data and RGB data through the MIPI CSI-2 interface.

The module has one video channel that can control the capture of data into a capture area of up to 4096 × 4096 pixels when scaling is off (2048 × 2048 pixels when scaling is on). It can also provide vertical and horizontal scaling of the data by up to three and two times, respectively.

For captured video data, the VIN provides a color space conversion function from YCbCr-422 to RGB or RGB to YCbCr-422 and a format conversion function from RGB to ARGB.

As the VIN internally generates a field signal, it can capture progressive data.

Input of data through the MIPI Alliance CSI-2 interfaces is possible.

Note: The upper limits on size clipping depend on the use case.

68.1.1 Features

Table 68.1 Functional overview of VIN (1 of 2)

Parameter	Function
Input interface	Data Type: YCbCr-422 8-bit data*1 YCbCr-422 10-bit data*1 *2 RGB-888 data*1 8-bit user defined data (RAW8)*1 *2
Internal Sync Signal Generation	Field signals usually refer to even and odd fields of interlaced signals, but in this VIN, they are used for progressive input. Field signals should be read as frame signals. VIN operates based on a field signal, it is necessary to have the field signal generated internally if there is no field signal. The field signal generation method differs between CSI-2 IF interlaced input and progressive input. ● For interlaced inputs in CSI-2 IF, the channel even field detection control register (CSIFLD.FLD_EN) should be used to generate the field signal. If the frame number in a frame start packet or frame end packet is detected and matches the CSIFLD .FLD_NUM register, the field signal is asserted to 1b (even field). ● For progressive inputs on CSI-2 IF, the channel even field detection control register should be off (CSIFLD.FLD_EN = 0). Toggle the field signal at each frame start packet assertion.
Capture Mode	The following three modes can be selected to capture the interlace images. Triple-buffering control is provided in accordance with the captured field image to coordinate with the video capture mode of the display module. ● Odd-field capture mode ● Even-/odd-field capture mode ● Even-field capture mode
Vertical and Horizontal Scaling	Factors for scaling up or down can be set to values in the range from one sixteenth to three in the vertical direction and from one sixteenth to two in the horizontal direction. When scaling is performed, the maximum size for both input and output is 2048 × 2048 pixels.
Size Clipping	VIN has a clipping circuit that processes a maximum of 4096 × 4096 pixels for the input image and a clipping circuit that processes a maximum of 2048 × 2048 pixels for the image after scaling. Each can be controlled independently.
Color Space Conversion	Color space conversion can be performed from YC to RGB or from RGB to YC. Desired conversion coefficients can be specified through registers to adjust colors.
Lookup Table (LUT) Precision Conversion	The lookup table (LUT) precision can be converted from 10 bits to 8 bits for each pixel according to the color space conversion result. The lookup table is common to the YCbCr and RGB formats.

Table 68.1 Functional overview of VIN (2 of 2)

Parameter	Function
Image Data Format Conversion	To the precision converted YCbCr422 or RGB888 image data, the following data format conversions are applicable. ● YCbCr image data – Y/Cb/Cr multiplex conversion – YCbCr422 → YC separation (separated into Y and CbCr components.) – YCbCr422 → Y component extraction ● RGB image data – RGB-888 → 32-bit/pixel conversion – RGB-888 → RGB-565 (16-bit/pixel) conversion – RGB-888 → ARGB-1555 (16-bit/pixel) conversion – RGB-888 → ARGB-8888 (32-bit/pixel) conversion
Memory Output Data Format	Format-converted image data can be transferred to the memory. The following shows the available formats of the data stored. ● YCbCr image data – Y/Cb/Cr422, 8-bit multiplexed – Y/Cb/Cr422, 10-bit multiplexed – YC separation, YCbCr422, Y, 8-bit Cb/Cr, 8-bit multiplexed – YC separation, YCbCr422, Y, 10-bit Cb/Cr, 8-bit multiplexed – YC separation, YCbCr422, Y, 10-bit Cb/Cr, 10-bit multiplexed – YC separation, Y data, 8-bit – YC separation, Y data, 10-bit ● RGB image data – RGB-565 (16 bits/pixel) – ARGB-1555 (16 bits/pixel) – RGB-888 (32 bits/pixel) – ARGB-8888 (32 bits/pixel) ● RAW data – RAW8 (8 bits/pixel)
VIN Operating clock	AXI bus clock (aclk) runs on ICLK. The registers operate with PCLKA of the peripheral bus clock. The following frequency ratios can be set. ICLK:PCLKA = 1:1, 2:1, 4:1 aclk is the main clock for VIN, and the circuit processes one clock and one pixel. In the case of CSI-2, VIN processes the following depths with aclk, so select a clock that is higher than the transfer rate. RGB 888: depth = 24 bit YCbCr8bit: depth = 16 bit YCbCr10bit: depth = 20 bit RAW8: depth = 16 bit Example: When setting up a 720 Mbps 2 lane with YCbCr 8bit scaling bypass, aclk must be set to a frequency of 720*2/16 = 90 MHz or higher. See section 68.3.4. Scaling.

Note 1. Inputs are in the formats which conform to the standard for the MIPI CSI-2 interface.
Note 2. Disable the XY scaling setting (only the 100% scaling is enabled).

See Figure 62.1 for the block diagram.

68.2 Register Description

Access the following registers only in units of 32 bits.

68.2.1 MC : Main Control Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	CLP[1:0]		—	SCLE	YUV444	DC2	—	ST	—	LUTE	—	INF[2:0]		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	DC[1:0]		—	—	—	—	—	—	—	EN	—	IM[1:0]		—	BPS	ME
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0

Bit	Symbol	Function	R/W
0	ME	Module Enable This is the enable bit for the VIN. Set this bit before setting the Frame Capture register (FC). 0: The module operation is stopped. 1: The module operation is enabled*3.	R/W
1	BPS	Color Space Conversion Bypass Mode YCbCr → RGB or RGB → YCbCr conversion is performed with the coefficients specified by the YC-RGB conversion coefficient register or RGBYC coefficient register. If the input format is RAW, set 1b. 0: The input YCbCr data is converted into the RGB color space and RGB data is converted into the YCbCr color space*2. 1: Color space conversion is not performed.	R/W
2	—	This bit is read as 0. The write value should be 0.	R/W
4:3	IM[1:0]	Interlace Mode These bits specify the capture mode. Do not modify this setting during capture operation. 0 0: Odd-field (field 1) capture mode Handles only odd fields as frames and stores them in external memory. 0 1: Odd-/even-field capture mode Handles odd and even fields as separate frames and stores them in external memory. This mode is available only in continuous frame capture mode. For progressive input and continuous frame capture mode, set 01b. 1 0: Even-field (field 2) capture mode Handles only even fields as frames and stores them in external memory. 1 1: Setting prohibited	R/W
5	—	This bit is read as 0. The write value should be 0.	R/W
6	EN	Endian Type This bit specifies the endian type for data to be output to external memory. When allocating the YCbCr422 (UYVY format) data in big endian, be sure to set the BPSM bit in DMR to 1. 0: Image data is packed and allocated in little endian. 1: Image data is packed and allocated in big endian.	R/W
13:7	—	These bits are read as 0. The write value should be 0.	R/W
15:14	DC[1:0]	Dithering Mode Control These bits select the dithering mode for conversion from RGB888 to RGB565/ARGB1555. 0 0: Dithering with cumulative addition 0 1: Ordered dithering 1 1 0: Setting prohibited 1 1: Ordered dithering 2	R/W

Bit	Symbol	Function	R/W
18:16	INF[2:0]	Input Interface Format These bits specify the format of images input to the VIN. 0 0 0: Setting prohibited 0 0 1: 8-bit YCbCr-422*1 0 1 0: Setting prohibited 0 1 1: 10-bit YCbCr-422*1 1 0 0: 8-bit user defined data (RAW8)*1 1 0 1: Setting prohibited 1 1 0: 24-bit RGB-888*1 1 1 1: Setting prohibited	R/W
19	—	This bit is read as 0. The write value should be 0.	R/W
20	LUTE	Lookup Table Enable This bit enables conversion from 10 bits to 8 bits through the lookup table (LUT). To perform LUT conversion, the conversion table should be set with LUTP and LUTD. 0: LUT is not used. 1: LUT is used.	R/W
21	—	This bit is read as 0. The write value should be 0.	R/W
22	ST	Initialization control at STartup Write 1b to this bit to initialize the internal state. 0: Invalid. 1: Perform the initialization procedure. It is prohibited to use it except for the initialization and resumption procedure.	W
23	—	This bit is read as 0. The write value should be 0.	R/W
24	DC2	Dithering mode Control 2 This bit is valid when MC.DC[1:0] is set to 11. This bit is used to enable or disable dithering in the frame direction. 0: Enables dithering in the frame direction 1: Disables dithering in the frame direction	R/W
25	YUV444	YUV444 conversion This bit selects how to convert YCbCr422 data to YCbCr444. 0: The data of CbCr to be interpolated holds the data one pixel before. 1: Generates CbCr data to be interpolated. This register is invalid when the input format is RGB. If the input format is RAW, set 0.	R/W
26	SCLE	This bit is used to enable or disable scaling by the UDS. 0: Disables scaling by the UDS. 1: Enables scaling by the UDS.	R/W
27	—	This bit is read as 0. The write value should be 0.	R/W
29:28	CLP[1:0]	Pixel Data Clipping	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note 1. Table 68.2 shows the combinations of interfaces that can be set by the input interface format (the INF bits in MC). Do not make any setting that is not listed in the table.

Note 2. Table 68.3 shows the image data that can be converted according to the color space conversion bypass mode settings (set by the BPS bit in MC).

Note 3. To stop capturing operation, only set the ME bit to 0. Do not change the other bit settings.

Table 68.2 Capture interface settings

Interface	MC/INF
YCbCr-422 8-bit data (MIPI CSI-2)	0 0 1
YCbCr-422 10-bit data (MIPI CSI-2)	0 1 1
RGB-888 data (MIPI CSI-2)	1 1 0
8-bit user defined data (RAW8) (MIPI CSI-2)	1 0 0

Table 68.3 Captured data formats

Input data format	MC/INF	MC/BPS	MC/IM	MC/YUV444	Captured data format
YCbCr-422 (MIPI CSI-2)	0 0 1/0 1 1	0	—	0 or 1	RGB format
		1	—	0 or 1	YCbCr format
RGB-888 data (MIPI CSI-2)	1 1 0	0	—	—	YCbCr format
		1	—	—	RGB format
8-bit user defined data (RAW8)	1 0 0	1	0 1	0	User defined format

CLP[1:0] bits (Pixel Data Clipping)

When the input image data is in the YCbCr format, these bits specify the data clip value for clipping the YCbCr-RGB color conversion input data to the nominal range prescribed in the ITU-R BT.601 standard.

Table 68.4

CLP	Luminance	Color Difference	
0 0	No clipping	16 or a smaller value is clipped to 16. 240 or a greater value is clipped to 240.	Initial condition
0 1	16 or a smaller value is clipped to 16. 240 or a greater value is clipped to 240.	16 or a smaller value is clipped to 16. 240 or a greater value is clipped to 240.	—
1 1	No clipping	No clipping	—

Note: The internal processing is done in 12bit, but it is written in 8bit.

68.2.2 MS : Module Status Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	FMS[1:0]		—		MA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	1	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	FBS[1:0]		FS	AV	CA
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	1	1	0	0

Bit	Symbol	Function	R/W
0	CA	Video Capture Active Status This bit shows the current video capture operation status. This bit is updated by the captured field signal. In field capture mode (MC.IM = 00b or 10b), this bit is set to 1 even in fields that do not output data to external memory. 0: Video capture is not operating. 1: Video capture is operating.	R
1	AV	Active Video Status This bit shows whether the current field is in the active video area defined by the pre-clipping register. This bit will be 0 if no input data is captured. 0: The current field is not in the active video area. 1: The current field is in the active video area.	R

Bit	Symbol	Function	R/W
2	FS	Field Status This bit shows the type of the current capture field. When video capture is operating, this bit should be read after the FIS bit in INTS is set to 1. 0: The current field is an odd field (field 1). 1: The current field is an even field (field 2).	R
4:3	FBS[1:0]	Frame Buffer Status These bits show the frame buffer status. When video capture is operating, these bits should be read after the FIS bit in INTS is set to 1. The following is set when the field signal is updated (switched). INTS.FIS is set by field switching. 0 0: The latest valid frame buffer has the base address defined by the memory base 1 register. 0 1: The latest valid frame buffer has the base address defined by the memory base 2 register. 1 0: The latest valid frame buffer has the base address defined by the memory base 3 register. 1 1: There is no valid frame buffer.	R
15:5	—	These bits are read as 0.	R
16	MA	External frame Memory capture Active status This bit is updated in external memory when the capture operation is complete (AXI's write response mbvalid is returned). In field capture mode (MC.IM [1:0] = 00b and 10b), this bit is set to 1b even in fields that do not output data to external memory. 0: Capture to external memory is stopped. 1: Capture to external memory is in operation.	R
18:17	—	These bits are read as 0.	R
20:19	FMS[1:0]	External Frame Memory buffer Status These bits indicate the status of the external frame buffer. The following confirms that the data has been captured in the external memory (the write response mbvalid of AXI has returned) and sets it. 0 0: The latest valid frame buffer has the base address defined by the memory base 1 register. 0 1: The latest valid frame buffer has the base address defined by the memory base 2 register. 1 0: The latest valid frame buffer has the base address defined by the memory base 3 register. 1 1: There is no valid frame buffer.	R
31:21	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

FMS[1:0] bits (External Frame Memory buffer Status)

When video capture is operating, these bits should be read after the FMS bit in the INTS is set to 1.

INTS.FMS confirms that the data has been captured in the external memory (the write response mbvalid of AXI has returned) and sets it.

68.2.3 FC : Frame Capture Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	CC	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	—	This bit is read as 0. The write value should be 0.	R/W
1	CC	Continuous Frame Capture Mode 0: The continuous frame capture mode is not set. 1: The continuous frame capture mode is set.	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

CC bit (Continuous Frame Capture Mode)

This bit specifies the continuous frame capture mode. In this mode, the first capture frame is written into the memory address specified by the memory base 1 register (MB1). After that, the capture operation is repeated in the order of MB2, MB3, MB1, and MB2.

Writing 0 into this bit during continuous capture operation will immediately terminate the capture operation if the current frame is completed or it has not been captured.

After the capture operation is resumed by writing 1 into this bit, the memory address where the first capture frame is written depends on the value indicated by the frame buffer status (FBS[1:0]) bits in the module status register (MS).

68.2.4 SLPRC : Start Line Pre-Clip Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x00C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	SLPRC[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	SLPRC[11:0]	Start Line PRe-Clip	R/W
31:12	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

SLPRC[11:0] bits (Start Line PRe-Clip)

These bits specify the (pre-clipping start line - 1) value in line units.

This value is used before scaling. Specify a value in the range of 0 to 4094 so that the number of lines after pre-clipping will be 2 or more.

Note:
- When SLPRC is set to 0, the first valid line is specified.
- Set the SLPRC value within vertical valid lines of the input.
- If scaling is used, ELPRC - SLPRC+1 must not exceed 2048 pixels.

68.2.5 ELPRC : End Line Pre-Clip Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	ELPRC[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	ELPRC[11:0]	End Line PRe-Clip	R/W
31:12	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

ELPRC[11:0] bits (End Line PRe-Clip)

These bits specify the (pre-clipping end line - 1) value in line units.

This value is used before scaling. Specify a value in the range of 1 to 4095 so that the number of lines after pre-clipping will be 2 or more.

Note:
- Set the ELPRC value within vertical valid lines of the input.
- When using vertical or horizontal scaling, specify a value within the range of 3 to 4095 so that the number of lines after pre-clipping will be 4 or more.
- When using scaling, ELPRC - SLPRC + 1 should not exceed 2048 pixels.
- If an interlaced image is being input, the value here should be such that "ELPRC-SLPRC+1" is half the original number of lines value.

68.2.6 SPPRC : Start Pixel Pre-Clip Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x014

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	SPPRC[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	SPPRC[11:0]	Start Pixel Pre-Clip	R/W
31:12	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

SPPRC[11:0] bits (Start Pixel Pre-Clip)

These bits specify the (pre-clipping start pixel - 1) value in pixel units.

This value is used before scaling. Specify a value in the range from 0 to 4090 so that the number of pixels after pre-clipping will be an even number larger than 6.

Note:
- When SPPRC is set to 0, the first horizontal valid pixel is specified.
- Set the SPPRC value within horizontal valid pixels of the input.
- When using scaling, EPPRC - SPPRC+1 must not exceed 2048 pixels.

68.2.7 EPPRC : End Pixel Pre-Clip Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x018

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	EPPRC[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	EPPRC[11:0]	End Pixel PRe-Clip	R/W
31:12	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

EPPRC[11:0] bits (End Pixel PRe-Clip)

These bits specify the (pre-clipping end pixel - 1) value in pixel units.

This value is used before scaling. Specify a value in the range of 5 to 4095 so that the number of pixels after pre-clipping will be an even number larger than 6.

Note:
- Set the EPPRC value within horizontal valid pixels of the input.
- When using scaling, EPPRC - SPPRC + 1 must not exceed 2048 pixels.

Table 68.5 Pre-clip area setting unit

Input format	Pre-clip start unit	Clipping size unit
YCbCr422 data	2 pixel	2 pixel
RGB data	2 pixel	2 pixel
RAW8 data	4 pixel	4 pixel

Figure 68.1 shows how to set the pre-clip.

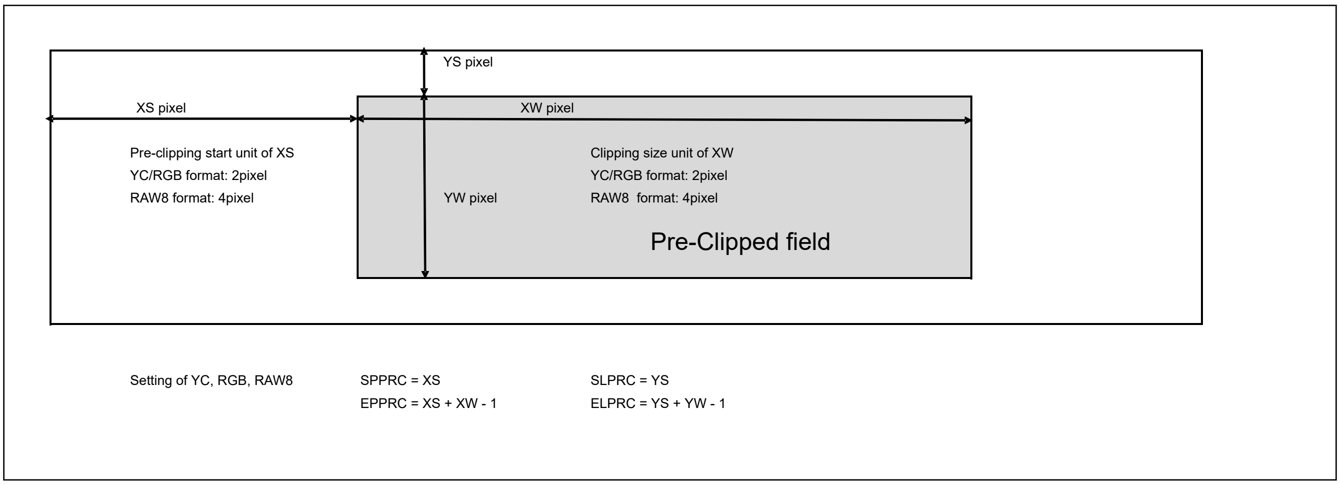


Figure 68.1 Pre-clip setting method

68.2.8 CSI_IFMD : CSI2 Interface Mode Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x020

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	DES0	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	DT[5:0]						—	—	—	—	VC_SEL[3:0]			
Value after reset:	0	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
3:0	VC_SEL[3:0]	Virtual Channel SELect This bit selects the CSI virtual channel to capture. 0 0 0 0: VC0 0 0 0 1: VC1 ⋮ 1 1 1 1: VC15	R/W
7:4	—	These bits are read as 0. The write value should be 0.	R/W
13:8	DT[5:0]	Data Type select Please select from the following. CSI-2 needs to output data to VIN with Generic CSI-2 rule. 0x1E: YUV422 8-bit 0x1F: YUV422 10-bit 0x24: RGB888 0x2A: RAW 8-bit	R/W
24:14	—	These bits are read as 0. The write value should be 0.	R/W
25	DES0	Data Extension Select This bit is used to select how data are expanded to 12 bits within the VIN module. 0: Empty bits in the input data are repeatedly expanded from the highest-order bit. 1: Empty bits will be padded with zeros.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

DES0 bit (Data Extension Select)

This bit must be set to 1 when the YCbCr-422/RAW interface is in use.

For example, in the case of R [7: 0], it will be as follows.

0b: {R [7: 0], R [7: 4]}

1b: {R [7: 0], 0000b}

68.2.9 CSIFLD : Field Detection Control Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x024

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	FLD_NUM
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	FLD_SEL[1:0]		—	—	—	FLD_EN
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0

Bit	Symbol	Function	R/W
0	FLD_EN	FieLD detect ENable Even field detection control. This bit turns detection of even fields on or off. If the input image is a progressive sequence, it should be set to 0. 0: Even field detection control disabled 1: Even field detection control enabled	R/W
3:1	—	These bits are read as 0. The write value should be 0.	R/W
5:4	FLD_SEL[1:0]	even FieLD DETect SELect 0 0: Setting is prohibited. 0 1: When FLD_NUM matches the field number [0] bit, the field is detected as an even field. 1 0: Setting is prohibited. 1 1: Setting is prohibited.	R/W
15:6	—	These bits are read as 0. The write value should be 0.	R/W
16	FLD_NUM	even FieLD NUMber setting This bit specifies a value for detecting even fields in the interlaced image. The field number of the frame start packet is referenced to detect even fields. If the field number [0] bit matches the setting of this bit, the field is detected as an even field.	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.10 IS : Image Stride Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x02C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	IS[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
12:0	IS[12:0]	Image Stride (Setting unit: pixel)	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

IS[12:0] bits (Image Stride (Setting unit: pixel))

These bits specify the image stride.

Specify a multiple of the setting unit for the image stride. (always write 0 to the lower 4 bits)

The image stride width on the memory is the number of bytes per "IS [12:0] × 1 pixel".

- When the UDS is in use:
  Specify a value no less than the post-clipping width (CL_HSIZE).
- When the UDS is not in use:
  - Other than RAW8
    Specify a value no less than the pre-clipping width (EPPRC - SPPRC) +1.
  - RAW8
    Specify a value no less than the pre-clipping width ((EPPRC - SPPRC) + 1)/2.

Table 68.6 Image stride setting unit

Output format	Setting unit (pixel)	In Image Stride Bytes per pixel
YCbCr422 8 bit	64	2
YCbCr422 10 bit	32	4
YC separation YCbCr422 Y (8 bit)/C (8 bit)	128	Y:1 C:1
YC separation YCbCr422 Y (10 bit)/C (10 bit)	64	Y:2 C:2
YC separation YCbCr422 Y (10 bit)/C (8 bit)	128	Y:2 C:1
RGB565	64	2
ARGB1555	64	2
RGB888	32	4
ARGB8888	32	4
RAW8[*1]	64	2

Note 1. RAW8 output is 1 byte/pixel, but the image stride is the number of pixels set in this bit × 2 bytes.

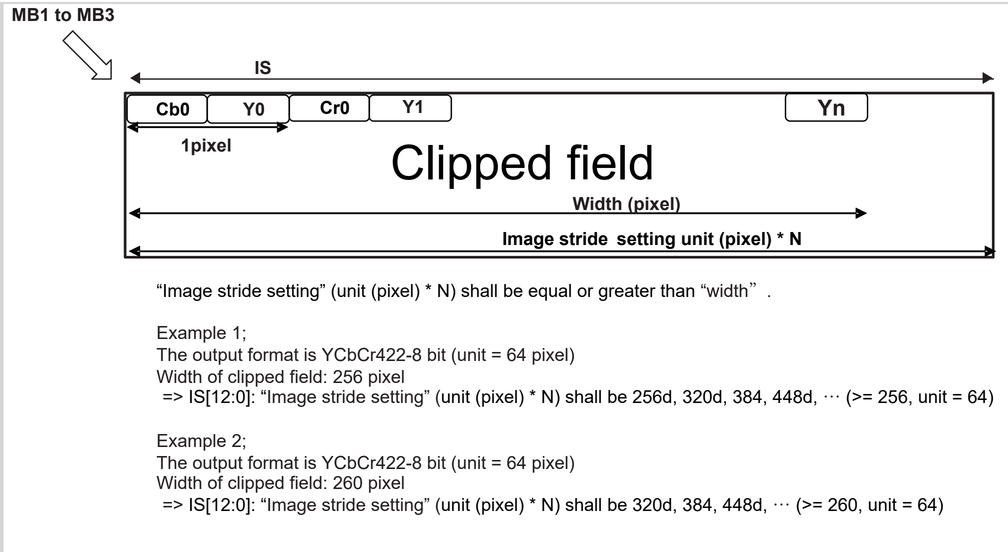


Figure 68.2 Image stride setting example memory base 1 register (MB1)

68.2.11 MB1 : Memory Base 1 Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x030

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MB1[24:9]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MB1[8:0]									—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	—	These bits are read as 0. The write value should be 0.*1	R/W
31:7	MB1[24:0]	Memory Base Address 1	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Reserved bits that indicate the lower-order seven bits of memory base address 1 (a multiple of 128 bytes).

MB1[24:0] bits (Memory Base Address 1)

These bits specify the transfer start address in frame buffer 1. Specify a value for physical address bits [31:7] in units of 128 bytes.

If the module is in continuous frame capture mode, this value is used as the MB1 address in the following capture sequence: MB1 → MB2 → MB3 → MB1 → MB2 → MB3.

Specify a memory address taking into account the image size so that the image data does not exceed an area boundary on the address map.

68.2.12 MB2 : Memory Base 2 Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x034

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MB2[24:9]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MB2[8:0]									—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	—	These bits are read as 0. The write value should be 0.*1	R/W
31:7	MB2[24:0]	Memory Base Address 2	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Reserved bits that indicate the lower-order seven bits of memory base address 2 (a multiple of 128 bytes).

MB2[24:0] bits (Memory Base Address 2)

These bits specify the transfer start address in frame buffer 2. Specify a value for physical address bits [31:7] in units of 128 bytes.

If the module is in continuous frame capture mode, this value is used as the MB2 address in the following capture sequence: MB1 → MB2 → MB3 → MB1 → MB2 → MB3.

Specify a memory address taking into account the image size so that the image data does not exceed an area boundary on the address map.

68.2.13 MB3 : Memory Base 3 Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x038

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	MB3[24:9]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	MB3[8:0]									—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	—	These bits are read as 0. The write value should be 0.*1	R/W
31:7	MB3[24:0]	Memory Base Address 3	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Reserved bits that indicate the lower-order seven bits of memory base address 3 (a multiple of 128 bytes).

MB3 bits (Memory Base Address 3)

These bits specify the transfer start address in frame buffer 3. Specify a value for physical address bits [31:7] in units of 128 bytes.

If the module is in continuous frame capture mode, this value is used as the MB3 address in the following capture sequence: MB1 → MB2 → MB3 → MB1 → MB2 → MB3.

Specify a memory address taking into account the image size so that the image data does not exceed an area boundary on the address map.

68.2.14 LC : Line Count Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x03C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	LC[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	LC[11:0]	Line Count These bits indicate the effective line position after PreClip of the current image input field. The first vertical valid line shows LC = 0x000. See Figure 68.3 for update timing.	R
31:12	—	These bits are read as 0.	R

Note: S-TYPE-3, P-TYPE-3

68.2.15 IE : Interrupt Enable Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x040

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	FIE2	—	—	—	—	—	—	—	—	—	—	—	—	—	VFE	VRE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	AREE	ROE	—	—	—	—	PRCLIPVEE	PRCLIPHEE	—	—	FME	FIE	—	SIE	EFE	FOE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FOE	FIFO Overflow Interrupt Enable 0: FIFO overflow interrupts are disabled. 1: FIFO overflow interrupts are enabled.	R/W
1	EFE	End of Frame Interrupt Enable 0: End of frame interrupts are disabled. 1: End of frame interrupts are enabled.	R/W
2	SIE	Scanline Interrupt Enable 0: Scanline interrupts are disabled. 1: Scanline interrupts are enabled.	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
4	FIE	Field Interrupt Enable This bit enables or disables field-switching interrupts. 0: Field-switching interrupts are disabled. 1: Field-switching interrupts are enabled.	R/W
5	FME	Frame Memory write completion interrupt Enable This bit enables or disables interrupts due to completion of writing to external memory. 0: Memory write completion interrupt disabled 1: Memory write completion interrupt enabled	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
8	PRCLIPHEE	PRCLIPH Error interrupt Enable This bit enables or disables PRCLIPH error detection interrupts. 0: PRCLIPH error detection interrupt disabled 1: PRCLIPH error detection interrupt enabled	R/W
9	PRCLIPVEE	PRCLIPV Error interrupt Enable This bit enables or disables PRCLIPV error detection interrupts. 0: PRCLIPV error detection interrupt disabled 1: PRCLIPV error detection interrupt enabled	R/W
13:10	—	These bits are read as 0. The write value should be 0.	R/W
14	ROE	Response Overflow interrupt Enable This bit enables or disables response overflow detection interrupts. 0: Response overflow detection interrupt disabled 1: Response overflow detection interrupt enabled	R/W
15	AREE	Axi Resp Error interrupt Enable This bit enables or disables AXI Resp error detection interrupts. 0: AXI Resp error detection interrupt disabled 1: AXI Resp error detection interrupt enabled	R/W
16	VRE	VSYNC Deasserting Detect Interrupt Enable This bit enables or disables VSYNC deasserting detect interrupts. 0: VSYNC deasserting detect interrupts are disabled. 1: VSYNC deasserting detect interrupts are enabled.	R/W
17	VFE	Vsync asserting detect interrupt Enable This bit enables or disables VSYNC asserting detect interrupts. 0: VSYNC asserting detect interrupts are disabled. 1: VSYNC asserting detect interrupts are enabled.	R/W
30:18	—	These bits are read as 0. The write value should be 0.	R/W
31	FIE2	Field Interrupt Enable 2 This bit enables or disables field interrupts 2. 0: Field interrupts are disabled. 1: Field interrupts are enabled.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.16 INTS : Interrupt Status Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x044

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	FIS2	—	—	—	—	—	—	—	—	—	—	—	—	—	VFS	VRS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ARES	ROS	—	—	—	—	PRCLIPVES	PRCLIPHES	—	—	FMS	FIS	—	SIS	EFS	FOS
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	FOS	FIFO Overflow Interrupt Status This bit shows that the FIFO has overflowed. After being set to 1, this bit is cleared to 0 by writing 1. Be sure to clear this bit to 0 before using it.	R/W
1	EFS	End of Frame Interrupt Status	R/W
2	SIS	Scanline Interrupt Status	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	FIS	Field Interrupt Status	R/W
5	FMS	Frame Memory write completion interrupt Status	R/W
7:6	—	These bits are read as 0. The write value should be 0.	R/W
8	PRCLIPHES	PRCLIPH Error interrupt Status	R/W
9	PRCLIPVES	PRCLIPV Error interrupt Status	R/W
13:10	—	These bits are read as 0. The write value should be 0.	R/W
14	ROS	Response Overflow interrupt Status	R/W
15	ARES	Axi Resp Error interrupt Status This bit shows that an AXI Resp error has been detected. After being set to 1, this bit is cleared to 0 by writing 1. Be sure to clear this bit to 0 before using it.	R/W
16	VRS	VSYNC Deasserting Detect Interrupt Status	R/W
17	VFS	VSYNC Asserting Detect Interrupt Status	R/W
30:18	—	These bits are read as 0. The write value should be 0.	R/W
31	FIS2	Field Interrupt Status 2 This bit shows that the field has changed. After being set to 1, this bit is cleared to 0 by writing 1. This bit is set to 1 regardless of whether a capture operation is in progress (regardless of MA, CA). Be sure to clear this bit to 0 before using it.	R/W

Note: S-TYPE-3, P-TYPE-3

EFS bit (End of Frame Interrupt Status)

This bit shows that the last frame has been reached in the active capture operation (MS.CA = 1).

This bit is set to 1 at the end of the even field (field 2).

After being set to 1, this bit is cleared to 0 by writing 1.

This bit is set to 1 only when a capture operation is in progress (MS.CA = 1).

Be sure to clear this bit to 0 before using it.

SIS bit (Scanline Interrupt Status)

This bit shows that the number of lines specified by SI has been reached in the active capture operation (MS.CA = 1). After being set to 1, this bit is cleared to 0 by writing 1.

This bit is set to 1 when the value of the LC register matches the set value of the SI register.

This timing is shown in Figure 68.3, Scanline Interrupt Status Generation Timing.

This bit is set to 1 only when a capture operation is in progress (MS.CA = 1).

Be sure to clear this bit to 0 before using it.

FIS bit (Field Interrupt Status)

This bit shows that a field has been captured in the active capture operation (MS.CA = 1). Even in fields that do not output data to external memory in field capture mode (MC.IM = 0 0 and 1 0), it is set by field switching.

After being set to 1, this bit is cleared to 0 by writing 1.

This bit is set to 1 only when a capture operation is in progress (MS.CA = 1).

Be sure to clear this bit to 0 before using it.

FMS bit (Frame Memory write completion interrupt Status)

This bit shows that a write to memory has completed during an external memory output operation (MS.MA = 1).

Confirm that the writing to the memory is completed after the field switching operation (AXI write response (mbvalid) is returned) and set it. Even in fields that do not output data to external memory in field capture mode (MC.IM = 0 0 and 1 0), it is set immediately after field switching.

After being set to 1, this bit is cleared to 0 by writing 1.

This bit is set to 1 only during capture to external memory (MS.MA = 1).

Be sure to clear this bit to 0 before using it.

PRCLIPHES bit (PRCLIPH Error interrupt Status)

This bit is set when the horizontal valid pixels of the input is smaller than the EPPRC-1 value in the PRCLIP area setting.

After being set to 1, this bit is cleared to 0 by writing 1. If this error is set, assert reset and return.

Be sure to clear this bit to 0 before using it.

PRCLIPVES bit (PRCLIPV Error interrupt Status)

This bit is set when the vertical valid lines of the input is smaller than the ELPRC-1 value in the PRCLIP area setting.

After being set to 1, this bit is cleared to 0 by writing 1. If this error is set, assert reset and return.

Be sure to clear this bit to 0 before using it.

ROS bit (Response Overflow interrupt Status)

This bit is set if the field transmission currently being captured does not complete writing to memory by the time the next field completes (the AXI write response mbvalid does not come back).

If this bit is set, the FMS (Memory Write Completion Interrupt) will not be set correctly.

Take measures such as increasing the bandwidth of AXI. After being set to 1, this bit is cleared to 0 by writing 1.

Be sure to clear this bit to 0 before using it.

VRS bit (VSYNC Deasserting Detect Interrupt Status)

This bit shows that a VSYNC deasserting has been detected.

After being set to 1, this bit is cleared to 0 by writing 1.

This bit is set to 1 regardless of whether a capture operation is in progress (regardless of MA, CA). Be sure to clear this bit to 0 before using it.

VFS bit (VSYNC Asserting Detect Interrupt Status)

This bit shows that a VSYNC asserting has been detected.

After being set to 1, this bit is cleared to 0 by writing 1.

This bit is set to 1 regardless of whether a capture operation is in progress (regardless of MA, CA). Be sure to clear this bit to 0 before using it.

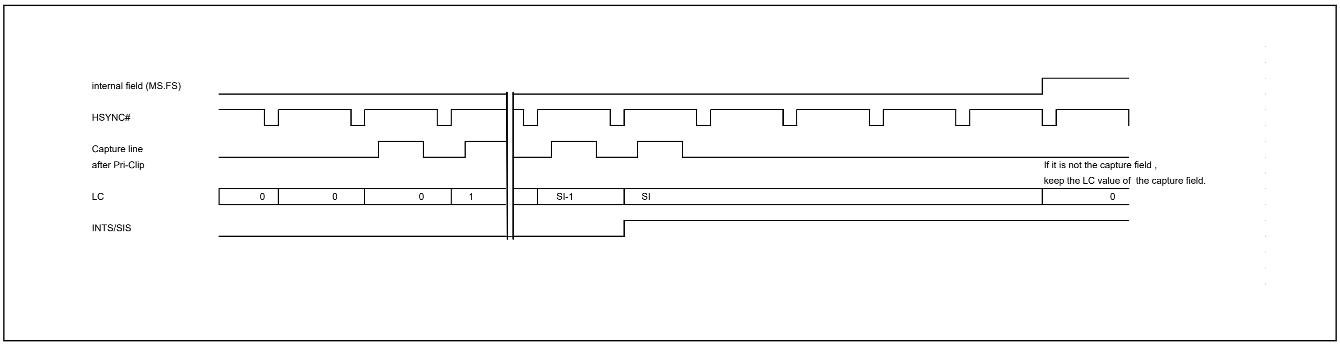


Figure 68.3 Scanline interrupt status generation timing

68.2.17 SI : Scanline Interrupt Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x048

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	SI[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	SI[11:0]	Scanline Interrupt Setting These bits specify a value to be compared with the LC register value in each field while the SIE bit in the IE register is set to 1. When this value matches the LC register value, an interrupt signal is asserted. When these bits are set to 0x000, the value of the scanline interrupt status bit (SIS) in the interrupt status register (INTS) is always 0.	R/W
31:12	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.18 MTCSTOP : AXI Transfer Stop Control Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x054

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	OUTSTAND[5:0]					
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	STOPACK	STOPREQ
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	STOPREQ	axi forced STOP REQuest Setting STOPREQ to 1, stops the issuance of new AXI requests, and if there is an AXI outstanding response, asserts STOPACK after receiving the response. After setting STOPREQ to 1 and checking STOPACK = 1, be sure to initialize by resetting. Asserting STOPREQ may assert the interrupt status INTS.FOS.	R/W
1	STOPACK	for axi forced STOP request, ACKnowledgement AXI forced stop request STOPREQ completion flag.	R
15:2	—	These bits are read as 0. The write value should be 0.	R/W
21:16	OUTSTAND[5:0]	OUTSTANDing current number Indicates the current number of outstandings. 0d is no outstanding.	R
31:22	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.19 DMR : Data Mode Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x058

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	A8BIT[7:0]								—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	YMODE[2:0]			YC_THR	—	—	EXRGB	—	—	—	BPSM	—	ABIT	DTMD[1:0]	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	DTMD[1:0]	Data Conversion Mode These bits set the format for storing RGB or YCbC data after LUT conversion, in the external memory.*1 0 0: Data is not converted. 0 1: RGB is converted to ARGB before output. 1 0: YC is separated before output.*2 1 1: Setting prohibited	R/W
2	ABIT	Alpha Bit This bit specifies the alpha value for data in ARGB-1555 output mode. 0: The alpha value is set to 0. 1: The alpha value is set to 1.	R/W
3	—	This bit is read as 0. The write value should be 0.	R/W
4	BPSM	Output Data Byte Swap Mode 0: Bytes are not swapped in output data. 1: Bytes are swapped in output data.	R/W
7:5	—	These bits are read as 0. The write value should be 0.	R/W
8	EXRGB	Extension RGB Conversion Mode 0: RGB data extension processing is not performed. 1: Data is extended to 32-bit RGB conversion when DTMD[1:0] is set to 00 or 01 as the data conversion mode.	R/W
10:9	—	These bits are read as 0. The write value should be 0.	R/W
11	YC_THR	YC Data Through Mode Y and CbCr data are transferred to memory as 10-bit data in accordance with the input format when this bit is set to 1. 0: Y and CbCr data are transferred to memory in accordance with the setting in the YMODE[2:0] bits. 1: Y and CbCr data are transferred to memory as 10-bit data in accordance with the input format.	R/W
14:12	YMODE[2:0]	YC Data Transfer Mode These bits specify the transfer method of Y/CbCr when the data conversion mode (DTMD) is 10 (YC separation). 0 0 0: Both Y and CbCr data are transferred to memory. 0 0 1: Only Y data is transferred to memory as 8-bit data. 0 1 0: 10-bit Y data and 8-bit CbCr data are transferred to memory 0 1 1: Only Y data is transferred to memory as 10-bit data 1 0 0: Setting prohibited 1 0 1: Setting prohibited 1 1 0: Setting prohibited 1 1 1: Setting prohibited	R/W
23:15	—	These bits are read as 0. The write value should be 0.	R/W
31:24	A8BIT[7:0]	Alpha 8 These bits set the alpha value for the ARGB8888 format output.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. The data conversion modes that can be set are shown in Table 68.7. Do not set any other mode that is not listed in the table. RGB and YCbCr data after LUT conversion should be set as shown in Table 68.7.
Note 2. Do not set for any other data format except for YCbCr422; set YC separation only for YCbCr data format.

BPSM bit (Output Data Byte Swap Mode)

To transfer in UYVY format with little endian, set BPSM to 0.

To transfer in YUYV format with little endian, set BPSM to 1.

To transfer in UYVY format with big endian, set BPSM to 1.

To transfer in YUYV format with big endian, set BPSM to 0.

Table 68.7 Data conversion settings (1 of 3)

RGB data conversion modes					
Format of data stored in memory	**DMR/ YMODE[2:0]**	**DMR/EXRGB**	**DMR/ YC_THR**	**DMR/DTMD [1:0]**	**Remarks**
RGB-565 (16 bits/pixel) format	0 0 0	0	0	0 0	—
RGB-888 (32 bits/pixel) format	0 0 0	1	0	0 0	—
ARGB-1555 (16 bits/pixel) format	0 0 0	0	0	0 1	The alpha bit is set with the ABIT bit in DMR.
ARGB-8888 (32 bits/pixel) format	0 0 0	1	0	0 1	The alpha bit is set with the A8BIT bit in DMR.

Table 68.7 Data conversion settings (2 of 3)

YCbCr data conversion modes					
Format of data stored in memory	**DMR/ YMODE[2:0]**	**DMR/EXRGB**	**DMR/ YC_THR**	**DMR/DTMD [1:0]**	**Remarks**
YCbCr-422 (8 bits) transfer	0 0 0	0	0	0 0	Little endian, to transfer in UYVY format, BPSM bit in DMR is set to 0; to transfer in YUYV format, BPSM bit in DMR is set to 1. Big-endian, to transfer in UYVY format, the BPSM bit in DMR is set to 1; to transfer in YUYV format, it is set to 0.
YCbCr-422 (10 bits) transfer	0 0 0	0	1	0 0	For little endian, transfer in UYVY format; for big endian, transfer in YUYV format.
Y (8 bits)/CbCr (8 bits) separation transfer	0 0 0	0	0	1 0	CbCr transfer destination is determined by the UVAOF register setting.
Y (10 bits)/CbCr (10 bits) separation transfer	0 0 0	0	1	1 0	CbCr transfer destination is determined by the UVAOF register setting.
Y (8 bits) transfer	0 0 1	0	0	1 0	—
Y (10 bits)/CbCr (8 bits) separation transfer	0 1 0	0	0	1 0	CbCr transfer destination is determined by the UVAOF register setting.
Y (10 bits) transfer	0 1 1	0	0	1 0	—

Table 68.7 Data conversion settings (3 of 3)

8-bit user defined data mode					
Format of data stored in memory	**DMR/ YMODE[2:0]**	**DMR/EXRGB**	**DMR/ YC_THR**	**DMR/DTMD [1:0]**	**Remarks**
RAW8	0 0 0	0	0	0 0	—

Note: If any combination of values not listed above is specified, correct operation is not guaranteed.

68.2.20 UVAOF : UV Address Offset Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x060

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	UVAOF[24:9]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	UVAOF[8:0]									—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	—	These bits are read as 0. The write value should be 0.*1	R/W
31:7	UVAOF[24:0]	UV Data Address Offset These bits specify the transfer offset address for the YC separation YCbCr-422 UV data. Specify bits 31 to 7 of the physical address in 128-byte units. The specified address should be equal to or greater than the Y transfer size. Otherwise, the overwriting of Y data occurs.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. Reserved bits that indicate the lower-order seven bits of UV data offset address (a multiple of 128 bytes).

68.2.21 YC-RGB Conversion Coefficient Registers

The YC to RGB color space conversion is performed using the following formula. Each coefficient can be set via a register.

R = CSCE1.YMUL2[13:0] × (Y- CSCE2.YSUB2 [11:0])

+ CSCE3.RCRMUL2 [13:0] × (Cr- CSCE2.CSUB2 [11:0])

G = CSCE1.YMUL2[13:0] × (Y- CSCE2.YSUB2 [11:0])

- CSCE3.GCRMUL2 [13:0] × (Cr- CSCE2.CSUB2 [11:0])

- CSCE4.GCBMUL2 [13:0] × (Cb- CSCE2.CSUB2 [11:0])

B = CSCE1.YMUL2[13:0] × (Y- CSCE2.YSUB2 [11:0])

+ CSCE4.BCBMUL2 [13:0] × (Cb- CSCE2.CSUB2 [11:0])

Note: The coefficients of ITU-R BT.601 (12 bits) are set as initial values.

$R = 1.164 \times (Y - 256) + 1.596 \times (Cr - 2048)$

$G = 1.164 \times (Y - 256) - 0.813 \times (Cr - 2048) - 0.392 \times (Cb - 2048)$

$B = 1.164 \times (Y - 256) + 2.017 \times (Cb - 2048)$

68.2.21.1 CSCE1 : YC to RGB Calculation Setting Extension Register 1

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x300

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ROUND
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	YMUL2[13:0]													
Value after reset:	0	0	0	1	0	0	1	0	1	0	0	1	1	1	1	1

Bit	Symbol	Function	R/W
13:0	YMUL2[13:0]	Y Multiplication Coefficient 2 for RGB Calculation These bits specify the multiplication coefficient for Y data in YCbCr → RGB color space conversion. (Initial value: 1.164) Specify an unsigned 14-bit integer obtained by multiplying the desired coefficient value by 4096.	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
16	ROUND	ROUND off enable This bit enables rounding for the calculation result in the YC → RGB color space conversion. Common to R, G, B. 0: Round down 1: Round off	R/W
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.21.2 CSCE2 : YC to RGB Calculation Setting Extension Register 2

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x304

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	YSUB2[11:0]											
Value after reset:	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	CSUB2[11:0]											
Value after reset:	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	CSUB2[11:0]	CbCr Subtraction Coefficient 2 for RGB Calculation These bits specify the subtraction coefficient for Cb and Cr data in YCbCr → RGB color space conversion. (Initial value: 2048) Do not change from the initial value . Specify an unsigned 12-bit integer.	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
27:16	YSUB2[11:0]	Y Subtraction Coefficient 2 for RGB Calculation These bits specify the subtraction coefficient for Y data in YCbCr → RGB color space conversion. (Initial value: 256) Specify an unsigned 12-bit integer.	R/W

Bit	Symbol	Function	R/W
31:28	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.21.3 CSCE3 : YC to RGB Calculation Setting Extension Register 3

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x308

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	RCRMUL2[13:0]													
Value after reset:	0	0	0	1	1	0	0	1	1	0	0	0	1	0	0	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	GCRMUL2[13:0]													
Value after reset:	0	0	0	0	1	1	0	1	0	0	0	0	0	0	1	0

Bit	Symbol	Function	R/W
13:0	GCRMUL2[13:0]	Cr Multiplication Coefficient 2 for G Calculation These bits specify the Cr multiplication coefficient for the G data calculation equation in YCbCr → RGB color space conversion. (Initial value: 0.813) Specify an unsigned 14-bit integer obtained by multiplying the desired coefficient value by 4096.	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
29:16	RCRMUL2[13:0]	Cr Multiplication Coefficient 2 for R Calculation These bits specify the Cr multiplication coefficient for the R data calculation equation in YCbCr → RGB color space conversion. (Initial value: 1.596) Specify an unsigned 14-bit integer obtained by multiplying the desired coefficient value by 4096.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.21.4 CSCE4 : YC to RGB Calculation Setting Extension Register 4

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x30C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	GCBMUL2[13:0]													
Value after reset:	0	0	0	0	0	1	1	0	0	1	0	0	0	1	0	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	BCBMUL2[13:0]													
Value after reset:	0	0	1	0	0	0	0	0	0	1	0	0	0	1	0	1

Bit	Symbol	Function	R/W
13:0	BCBMUL2[13:0]	Cb Multiplication Coefficient 2 for B Calculation These bits specify the Cb multiplication coefficient for the B data calculation equation in YCbCr → RGB color space conversion. (Initial value: 2.017) Specify an unsigned 14-bit integer obtained by multiplying the desired coefficient value by 4096.	R/W

Bit	Symbol	Function	R/W
15:14	—	These bits are read as 0. The write value should be 0.	R/W
29:16	GCBMUL2[13:0]	Cb Multiplication Coefficient 2 for G Calculation These bits specify the Cb multiplication coefficient for the G data calculation equation in YCbCr → RGB color space conversion. (Initial value: 0.392) Specify an unsigned 14-bit integer obtained by multiplying the desired coefficient value by 4096.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.22 UDS Control Registers

Note: The following operators, notations are defined for explaining the functions of UDS.

< x >: Discard decimal places of value x

68.2.22.1 UDS_CTRL : Scaling Control Registers

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x080

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	AMD	—	BLADV	—	—	—	—	—	—	—	BC	—	NE_RCR	NE_GY	NE_BCB
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	—	These bits are read as 0. The write value should be 0.	R/W
16	NE_BCB	B/Cb Interpolation Method When Bilinear/Nearest Neighbor Interpolation is Selected Specifies the interpolation method of the B/Cb component when bilinear/nearest neighbor interpolation is selected (BC = 0). 0: Bilinear method 1: Nearest neighbor method[*1]	R/W
17	NE_GY	G/Y Interpolation Method When Bilinear/Nearest Neighbor Interpolation is Selected Specifies the interpolation method of the G/Y component when bilinear/nearest neighbor interpolation is selected (BC = 0). 0: Bilinear method 1: Nearest neighbor method[*1]	R/W
18	NE_RCR	R/Cr Interpolation Method When Bilinear/Nearest Neighbor Interpolation is Selected Specifies the interpolation method of the R/Cr component when bilinear/nearest neighbor interpolation is selected (BC = 0). 0: Bilinear method 1: Nearest neighbor method[*1]	R/W
19	—	This bit is read as 0. The write value should be 0.	R/W
20	BC	Pixel Component Interpolation Method at Scale-Up/Down Specifies the method for interpolating pixel components at scale-up/-down. 0: Bilinear or nearest neighbor interpolation method is used 1: Interpolation method equivalent to 4 to 17 taps in accordance with the scaling factor is used (multi-tap mode)	R/W
27:21	—	These bits are read as 0. The write value should be 0.	R/W
28	BLADV	BiLinear or nearest neighbor interpolation characteristic ADVanced mode	R/W
29	—	This bit is read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
30	AMD	Advanced MoDe: Pixel Count at Scale-Up Specifies the number of pixels generated through scale-up in the UDS. This bit setting is ignored for scale-down. n: Number of pixels input to UDS 0: Pixel count after scale-up is 1 + < (n - 1) × scale-up factor> 1: Pixel count after scale-up is <n × scale-up factor>	R/W
31	—	This bit is read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note 1. This method can be used only when the scale-up/-down factor is 1/1 to 1/4.

BLADV bit (BiLinear or nearest neighbor interpolation characteristic ADVanced mode)

Controls the characteristics when bilinear or nearest neighbor interpolation.

Setting this bit to 1, improves aliasing characteristics at x1/2 to 1/8 scaling.

Note that the resolution around the edges may be reduced.

In multi-tap mode (BC = 1), this control is not available; to use this control, set the BC bit to 0.

68.2.22.2 UDS_SCALE : Scaling Factor Registers

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x084

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	HMANT[3:0]				HFRAC[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	VMANT[3:0]				VFRAC[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	VFRAC[11:0]	Multiplier (Fractional Part) of Vertical Scaling Factor These bits specify the fractional part of the vertical scaling factor. The image size to be obtained changes according to this setting. Calculate an appropriate value using the formula shown later to obtain a desired image size. A value from 0x556 to 0xFFF can be specified when an image is upscaled (the VMANT value is 0) in the vertical direction. A value from 0x000 to 0xFFF can be specified when an image is downscaled (the VMANT value is not 0) in the vertical direction. Select a value within the range shown in Table 68.9.	R/W
15:12	VMANT[3:0]	Multiplier (Integral Part) of Vertical Scaling Factor These bits specify the integral part of the vertical scaling factor. The image size to be obtained changes according to this setting. Calculate an appropriate value using the formula shown later to obtain a desired image size. A value from 0x0 to 0xF can be specified. Select a value within the range shown in Table 68.9.	R/W
27:16	HFRAC[11:0]	Multiplier (Fractional Part) of Horizontal Scaling Factor These bits specify the fractional part of the horizontal scaling factor. The image size to be obtained changes according to this setting. Calculate an appropriate value using the formula shown later to obtain a desired image size. A value from 0x800 to 0xFFF can be specified when an image is upscaled (the HMANT value is 0) in the horizontal direction. A value from 0x000 to 0xFFF can be specified when an image is downscaled (the HMANT value is not 0) in the horizontal direction. Select a value within the range shown in Table 68.8.	R/W

Bit	Symbol	Function	R/W
31:28	HMANT[3:0]	Multiplier (Integral Part) of Horizontal Scaling Factor These bits specify the integral part of the horizontal scaling factor. The image size to be obtained changes according to this setting. Calculate an appropriate value using the formula shown later to obtain a desired image size. A value from 0x0 to 0xF can be specified. Select a value within the range shown in Table 68.8.	R/W

Note: S-TYPE-3, P-TYPE-3

The HMANT and HFRAC bits set the scale-up/-down factor for an image in the horizontal direction, and the VMANT and VFRAC bits set the scale-up/-down factor for an image in the vertical direction. The UDS operation switches between horizontal scale-up and horizontal scale-down according to the HMANT and HFRAC bit settings, as shown in Table 68.8. (Setting a value outside the ranges of Table 68.8 in UDS operation is prohibited). Table 68.8 and Table 68.9 show the settings in the horizontal and vertical directions.

Note that settings for scaling in the horizontal direction and settings for scaling in the vertical direction can be made independently. Therefore, a setting for scale-up in the horizontal direction and scale-down in the vertical direction is possible. In such a case, because the formula (described later) for obtaining the image size after scale-up/-down is different between scale-up and scale-down, a formula matching the scale-up or scale-down operation should be selected independently for the horizontal direction and vertical direction.

Table 68.8 Switching of horizontal scale-up/down operation according to HMANT and HFRAC bit settings

HMANT	HFRAC	UDS operation
0x0	0x800 to 0xFFF	Scale-up
0x1	0x000	Same size (no scale-up/-down)
	0x001 to 0xFFF	Scale-down
0x2 to 0xF	0x000 to 0xFFF	Scale-down

Table 68.9 Switching of vertical scale-up/down operation according to VMANT and VFRAC bit settings

VMANT	VFRAC	UDS operation
0x0	0x556 to 0xFFF	Scale-up
0x1	0x000	Same size (no scale-up/-down)
	0x001 to 0xFFF	Scale-down
0x2 to 0xF	0x000 to 0xFFF	Scale-down

Described here is the method for calculating the size of the upscaled/downscaled image that was obtained based on this register setting. First, define the variables necessary for calculating the horizontal size of the upscaled/downscaled image, as shown below.

$$\text{hscale} = \frac{4096}{4096 \times m_h + f_h}$$

m_h is the value of UDS_SCALE.HMANT, and f_h is the value of UDS_SCALE.HFRAC.

This formula expresses the estimate of the scale-up/-down factor processed by the UDS. If the horizontal size of the image before scale-up/-down is set as hsize_{org}, the horizontal size of the image after scale-up/-down can be roughly obtained through $\text{hsize}_{org} \times \text{hscale}$.

Similarly, define the variables necessary for calculating the vertical size of the upscaled/downscaled image.

$$\text{vscale} = \frac{4096}{4096 \times m_v + f_v}$$

When setting mv as the value of UDS_SCALE.VMANT, fv as the value of UDS_SCALE.VFRAC, and vsize_{org} as the vertical size of the image before scale-up/-down, the vertical size of the image after scale-up/-down can be roughly obtained through $\text{vsize}_{org} \times \text{vscale}$.

When the UDS performs scale-down with the settings of Table 68.8, using the variables defined so far, the horizontal size of the downscaled image hsizedown_scaled and the vertical size of the downscaled image vsizedown_scaled become as follows:

$$\mathrm{hsize}_{\mathrm{down\_scaled}} = \left\langle 1 + \left(\left\langle 1 + \frac{\langle \mathrm{hsize}_{\mathrm{org}} - 1 \rangle}{m_h{}'} \right\rangle - 1 \right) \times m_h{}' \times \mathrm{hscale} \right\rangle$$

$$\mathrm{vsize}_{\mathrm{down\_scaled}} = \left\langle 1 + \left(\left\langle 1 + \frac{\langle \mathrm{vsize}_{\mathrm{org}} - 1 \rangle}{m_v{}'} \right\rangle - 1 \right) \times m_v{}' \times \mathrm{vscale} \right\rangle$$

Since the division of hscale and vscale has to be executed at the end of the above formulas, respectively, formulas considering the order of operation become as follows:

$$\mathrm{hsize}_{\mathrm{down\_scaled}} = \left\langle 1 + \left(\left(\left\langle 1 + \frac{\langle \mathrm{hsize}_{\mathrm{org}} - 1 \rangle}{m_h{}'} \right\rangle - 1 \right) \times m_h{}' \times 4096 \right) / \left(4096 \times m_h + f_h \right) \right\rangle$$

$$\mathrm{vsize}_{\mathrm{down\_scaled}} = \left\langle 1 + \left(\left(\left\langle 1 + \frac{\langle \mathrm{vsize}_{\mathrm{org}} - 1 \rangle}{m_v{}'} \right\rangle - 1 \right) \times m_v{}' \times 4096 \right) / \left(4096 \times m_v + f_v \right) \right\rangle$$

The value of m_h’ or m_v’, which is shown in Table 68.10, changes according to the setting of UDS_SCALE.HMANT or UDS_SCALE.VMANT.

Table 68.10 m_h’ or m_v’ setting

UDS_SCALE.HMANT Setting (UDS_SCALE.VMANT Setting)	m_h’ (m_v’)
1 to 3	1
4 to 7	2
8 to 15	4

When the UDS performs scale-up with the settings of Table 68.8 and UDS_CTRL.AMD is 0, the horizontal size of the upscaled image $\mathrm{hsize}_{\mathrm{up\_scaled}}$ and the vertical size of the upscaled image $\mathrm{vsize}_{\mathrm{up\_scaled}}$ become as follows:

$$\mathrm{hsize}_{\mathrm{up\_scaled}} = \left\langle 1 + \left(\mathrm{hsize}_{\mathrm{org}} - 1 \right) \times \mathrm{hscale} \right\rangle$$

$$\mathrm{vsize}_{\mathrm{up\_scaled}} = \left\langle 1 + \left(\mathrm{vsize}_{\mathrm{org}} - 1 \right) \times \mathrm{vscale} \right\rangle$$

Similar to the scale-down case, formulas considering the division of hscale and vscale become as follows:

$$\mathrm{hsize}_{\mathrm{up\_scaled}} = \left\langle 1 + \left(\left(\mathrm{hsize}_{\mathrm{org}} - 1 \right) \times 4096 \right) / \left(4096 \times m_h + f_h \right) \right\rangle$$

$$\mathrm{vsize}_{\mathrm{up\_scaled}} = \left\langle 1 + \left(\left(\mathrm{vsize}_{\mathrm{org}} - 1 \right) \times 4096 \right) / \left(4096 \times m_v + f_v \right) \right\rangle$$

When UDS_CTRL.AMD is 1, the horizontal size of the upscaled image hsizeup_scaled and the vertical size of the upscaled image vsizeup_scaled become as follows:

$$\mathrm{hsize}_{\mathrm{up\_scaled}} = \left\langle \mathrm{hsize}_{\mathrm{org}} \times \mathrm{hscale} \right\rangle$$

$$\mathrm{vsize}_{\mathrm{up\_scaled}} = \left\langle \mathrm{vsize}_{\mathrm{org}} \times \mathrm{vscale} \right\rangle$$

After considering the division of hscale and vscale, the formulas become as follows:

$$\mathrm{hsize}_{\mathrm{up\_scaled}} = \left\langle \left(\mathrm{hsize}_{\mathrm{org}} \times 4096 \right) / \left(4096 \times m_h + f_h \right) \right\rangle$$

$$\mathrm{vsize}_{\mathrm{up\_scaled}} = \left\langle \left(\mathrm{vsize}_{\mathrm{org}} \times 4096 \right) / \left(4096 \times m_v + f_v \right) \right\rangle$$

68.2.22.3 UDS_PASS_BWIDTH : Passband Registers

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x090

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	BWIDTH_H[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	BWIDTH_V[6:0]						
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
6:0	BWIDTH_V[6:0]	Vertical Signal Passband at Image Scale-Up/Down Set these bits following the method described later.	R/W
15:7	—	These bits are read as 0. The write value should be 0.	R/W
22:16	BWIDTH_H[6:0]	Horizontal Signal Passband at Image Scale-Up/Down Set these bits following the method described later.	R/W
31:23	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

The method for setting the passband in the horizontal direction of an image is described. When the UDS_SCALE.HMANT bits for horizontal scale-up/-down factor setting are not 0, set the BWIDTH_H bits according to the following formula. When the UDS_SCALE.HMANT bits are 0, set 64 in the BWIDTH_H bits.

$$\mathrm{BWIDTH\_H} = \left(64 \times \frac{4096 \times m_h'}{4096 \times m_h + f_h}\right)\left(\mathrm{VnUDS\_SCALE.HMANT} \neq 0\right)$$

$$\mathrm{BWIDTH\_H} = 64(\mathrm{VnUDS\_SCALE.HMANT} = 0)$$

m_h is the value of UDS_SCALE.HMANT, and f_h is the value of UDS_SCALE.HFRAC. For the m_h' value, see Table 68.10. The method for setting the passband in the vertical direction of an image is similar to that for the horizontal direction described earlier. Since only the correspondence relationship of the registers is changed as shown below, replace the variables in the previous explanation as shown below when reading.

BWIDTH_H → BWIDTH_V

$m_h' \rightarrow m_v'$

$m_h \rightarrow m_v$

$f_h \rightarrow f_v$

UDS_SCALE.HMANT → UDS_SCALE.VMANT

UDS_SCALE.HFRAC → UDS_SCALE.VFRAC

68.2.22.4 UDS_CLIP_SIZE : UDS Output Size Clipping Registers

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x0A4

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	CL_HSIZE[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	CL_VSIZE[11:0]											
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	CL_VSIZE[11:0]	Clipping Size of Vertical Pixel Count after Scale-Up/-Down CL_VSIZE specify the vertical size (vsize scaled) of the image actually output from the scaling filter. The setting range is 4 to 2,048 in a scale-up/down operation (see Table 68.8). These bits always have to be set when using the UDS, regardless of the scale-up, scale-down, or no-scaling setting by the UDS_SCALE register.	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
27:16	CL_HSIZE[11:0]	Clipping Size of Horizontal Pixel Count after Scale-Up/-Down CL_HSIZE specify the horizontal size ($hsize_{scaled}$) of the image actually output from the scaling filter. If the horizontal size ($hsize_{scaled}$) has an odd value, set CL_HSIZE to ($hsize_{scaled}$) -1. The setting range is 4 to 2,048 in a scale-up/down operation (see Table 68.8). These bits always have to be set when using the UDS, regardless of the scale-up, scale-down, or no-scaling setting by the UDS_SCALE register.	R/W
31:28	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Figure 68.4 shows the configuration of the UDS. The UDS consists of a scaling filter and clipping circuit, such as the configuration shown in Figure 68.4. The scaling filter and clipping circuit are independent of each other.

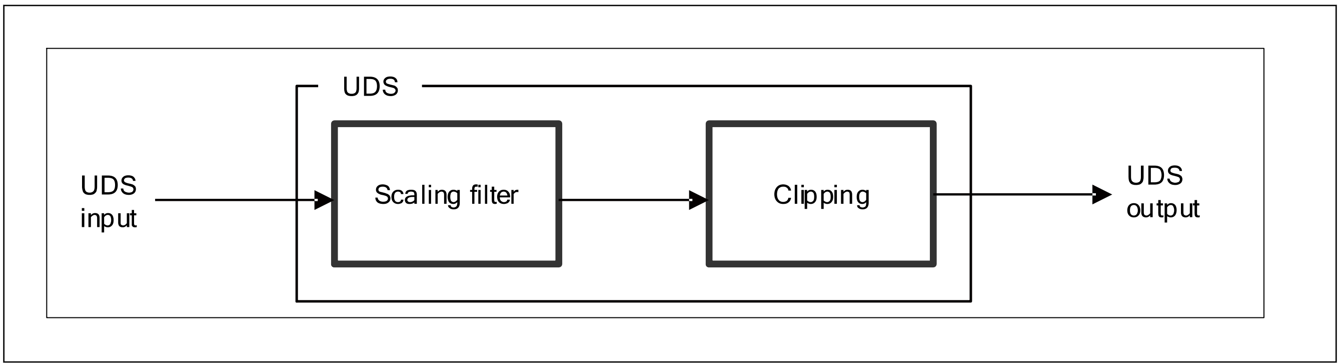


Figure 68.4 UDS configuration

The size of the image actually output by the scaling filter (see section 68.2.22.2. UDS_SCALE : Scaling Factor Registers for calculation) is determined from the size of the image input to the scaling filter and the UDS_SCALE setting;

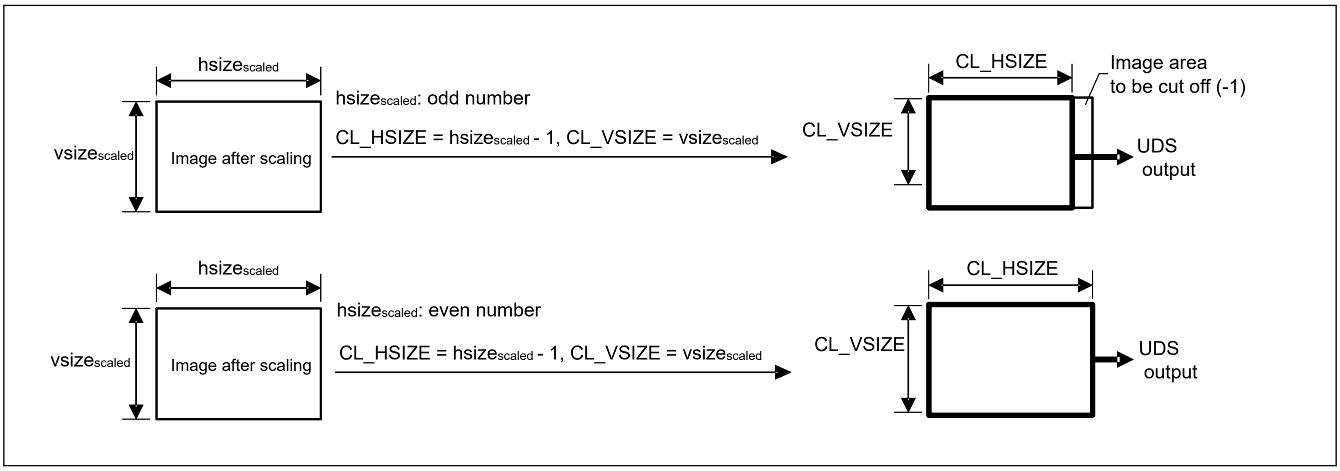


Figure 68.5 UDS output image for each CL_HSIZE/VSIZE setting

CL_HSIZE and CL_VSIZE should be set to the size of the image actually output from the scaling filter (hsize $_{scaled}$, vsize $_{scaled}$). If the horizontal size (hsize $_{scaled}$) takes an odd value, CL_HSIZE should be set to (hsize $_{scaled}$)-1. (Figure 68.5)

Do not set the CL_HSIZE and CL_VSIZE bits to values that satisfy CL_HSIZE > hsize$_{scaled}$ or CL_VSIZE > vsize$_{scaled}$.

68.2.23 LUTP : Lookup Table Pointer Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x100

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	LTYPR[9:0]										LTCBPR[9:0]			
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LTCBPR[9:0]						LTCRPR[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	LTCRPR[9:0]	Lookup Table Cr Pointer These bits set the LUT pointer to the Cr and B data obtained as the result of color space conversion.	R/W
19:10	LTCBPR[9:0]	Lookup Table Cb Pointer These bits set the LUT pointer to the Cb and G data obtained as the result of color space conversion.	R/W
29:20	LTYPR[9:0]	Lookup Table Y Pointer These bits set the LUT pointer to the Y and R data obtained as the result of color space conversion.	R/W
31:30	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

Note: Set the LUT pointer corresponding to the upper 10 bits of the 12-bit data of color space conversion result.
The access pointer to the LUT will be automatically incremented by writing to the LUTD register.
There will be no automatic increment at read access.

68.2.24 LUTD : Lookup Table Data Register

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x104

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	LTYDT[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	x	x	x	x	x	x	x	x

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	LTCBDT[7:0]								LTCRDT[7:0]							
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x

Bit	Symbol	Function	R/W
7:0	LTCRDT[7:0]	Lookup Table Cr Data These bits set the LUT conversion data for Cr and B data after color space conversion.	R/W
15:8	LTCBDT[7:0]	Lookup Table Cb Data These bits set the LUT conversion data for Cb and G data after color space conversion.	R/W
23:16	LTYDT[7:0]	Lookup Table Y Data These bits set the LUT conversion data for Y and R data after color space conversion.	R/W
31:24	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3
Note: The 8-bit data after LUT conversion is subjected to upper shift to carry out capture control as 12-bit data.

68.2.25 RGB-YC Conversion Coefficient Registers

Color space conversion from RGB to YC is done with the following formula. Each of the coefficients can be set with the registers.

Y = YCLRP × R + YCLGP × G + YCLBP × B + YCLAP

Cb = CBCLRP × R + CBCLGP × G + CBCLBP × B + CBCLAP

Cr = CRCLRP × R + CRCLGP × G + CRCLBP × B + CRCLAP

Note: Set the coefficients for ITU-R BT.601 (8 bits) as the initial value.

$$Y = 0.257 \times R + 0.504 \times G + 0.098 \times B + 256$$
$$Cb = -0.148 \times R - 0.291 \times G + 0.439 \times B + 2048$$
$$Cr = 0.439 \times R - 0.368 \times G - 0.071 \times B + 2048$$

68.2.25.1 YCCR1 : RGB to Y Calculation Setting Register 1

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x228

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	YCLRP[12:0]												
Value after reset:	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1	1

Bit	Symbol	Function	R/W
12:0	YCLRP[12:0]	R Multiplication Coefficient for Y Calculation These bits specify the R multiplication coefficient for the Y data calculation equation in RGB → YCbCr color space conversion. (Initial value: 263) Specify a 13-bit signed integer obtained by multiplying the desired coefficient value by 2^ YCLSFT. The MSB is a sign bit.	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.25.2 YCCR2 : RGB to Y Calculation Setting Register 2

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x22C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	YCLBP[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	1	1	0	0	1	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	YCLGP[12:0]												
Value after reset:	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0

Bit	Symbol	Function	R/W
12:0	YCLGP[12:0]	G Multiplication Coefficient for Y Calculation These bits specify the G multiplication coefficient for the Y data calculation equation in RGB → YCbCr color space conversion. (Initial value: 516) Specify a 13-bit signed integer obtained by multiplying the desired coefficient value by 2^ YCLSFT. The MSB is a sign bit.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
28:16	YCLBP[12:0]	B Multiplication Coefficient for Y Calculation These bits specify the B multiplication coefficient for the Y data calculation equation in RGB → YCbCr color space conversion. (Initial value: 100) Specify a 13-bit signed integer obtained by multiplying the desired coefficient value by 2^ YCLSFT. The MSB is a sign bit.	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.25.3 YCCR3 : RGB to Y Calculation Setting Register 3

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x230

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	YCLSFT[4:0]					YCLHEN	—	—	—	—	—	—	—
Value after reset:	1	0	0	0	1	0	1	0	0	0	0	0	0	0	0	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	YCLAP[11:0]											
Value after reset:	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	YCLAP[11:0]	Y Calculation Data Normalized Additional Value These bits set the Y data addition constant for RGB → YCbCr color space conversion. (Initial value: 256)	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
16	—	This bit is read as 1. The write value should be 1.	R/W
22:17	—	These bits are read as 0. The write value should be 0.	R/W
23	YCLHEN	Y Calculation Shift Down Result Round-Off Enable This bit enables round-off process for Y data calculation in RGB → YCbCr color space conversion 0: Round down to down shift process 1: Round-off to down shift process is enabled.	R/W
28:24	YCLSFT[4:0]	Y Calculation Shift Down Volume These bits set the amount of down shift for Y calculation in RGB → YCbCr color space conversion (Initial value: 10) Unit: Bit shift count Do not change from the initial value.	R/W
30:29	—	These bits are read as 0. The write value should be 0.	R/W
31	—	This bit is read as 1. The write value should be 1.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.25.4 CBCCR1 : RGB to Cb Calculation Setting Register 1

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x234

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	CBCLRP[12:0]												
Value after reset:	0	0	0	1	1	1	1	1	0	1	1	0	1	0	0	0

Bit	Symbol	Function	R/W
12:0	CBCLRP[12:0]	R Multiplication Coefficient for Cb Calculation These bits specify the R multiplication coefficient for the Cb data calculation equation in RGB → YCbCr color space conversion. (Initial value: -152) Specify a 13-bit signed integer obtained by multiplying the desired coefficient value by 2^ CBCLSFT. The MSB is a sign bit.	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.25.5 CBCCR2 : RGB to Cb Calculation Setting Register 2

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x238

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	CBCLBP[12:0]												
Value after reset:	0	0	0	0	0	0	0	1	1	1	0	0	0	0	1	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	CBCLGP[12:0]												
Value after reset:	0	0	0	1	1	1	1	0	1	1	0	1	0	1	1	0

Bit	Symbol	Function	R/W
12:0	CBCLGP[12:0]	G Multiplication Coefficient for Cb Calculation These bits specify the G multiplication coefficient for the Cb data calculation equation in RGB → YCbCr color space conversion. (Initial value: -298) Specify a 13-bit signed integer obtained by multiplying the desired coefficient value by 2^ CBCLSFT. The MSB is a sign bit.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
28:16	CBCLBP[12:0]	B Multiplication Coefficient for Cb Calculation These bits specify the B multiplication coefficient for the Cb data calculation equation in RGB → YCbCr color space conversion. (Initial value: 450) Specify a 13-bit signed integer obtained by multiplying the desired coefficient value by 2^ CBCLSFT. The MSB is a sign bit.	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.25.6 CBCCR3 : RGB to Cb Calculation Setting Register 3

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x23C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	CBCLSFT[4:0]					CBCLHEN	—	—	—	—	—	—	—
Value after reset:	1	0	0	0	1	0	1	0	0	0	0	0	0	0	0	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	CBCLAP[11:0]											
Value after reset:	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	CBCLAP[11:0]	Cb Calculation Data Normalized Additional Value These bits set the Cb data addition constant for RGB → YCbCr color space conversion. (Initial value: 2048) Do not change from the initial value.	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
16	—	This bit is read as 1. The write value should be 1.	R/W
22:17	—	These bits are read as 0. The write value should be 0.	R/W

Bit	Symbol	Function	R/W
23	CBCLHEN	Cb Calculation Shift Down Result Round-Off Enable This bit enables round-off process for Cb data calculation in RGB → YCbCr color space conversion 0: Round down to down shift process 1: Round-off to down shift process is enabled.	R/W
28:24	CBCLSFT[4:0]	Cb Calculation Shift Down Volume These bits set the amount of down shift for Cb calculation in RGB → YCbCr color space conversion (Initial value: 10) Unit: Bit shift count Do not change from the initial value.	R/W
30:29	—	These bits are read as 0. The write value should be 0.	R/W
31	—	This bit is read as 1. The write value should be 1.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.25.7 CRCCR1 : RGB to Cr Calculation Setting Register 1

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x240

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	CRCLRP[12:0]												
Value after reset:	0	0	0	0	0	0	0	1	1	1	0	0	0	0	1	0

Bit	Symbol	Function	R/W
12:0	CRCLRP[12:0]	R Multiplication Coefficient for Cr Calculation These bits specify the R multiplication coefficient for the Cr data calculation equation in RGB → YCbCr color space conversion. (Initial value: 450) Specify a 13-bit signed integer obtained by multiplying the desired coefficient value by 2^ CRCLSFT. The MSB is a sign bit.	R/W
31:13	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.25.8 CRCCR2 : RGB to Cr Calculation Setting Register 2

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x244

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	CRCLBP[12:0]												
Value after reset:	0	0	0	1	1	1	1	1	1	0	1	1	0	1	1	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	CRCLGP[12:0]												
Value after reset:	0	0	0	1	1	1	1	0	1	0	0	0	0	1	1	1

Bit	Symbol	Function	R/W
12:0	CRCLGP[12:0]	G Multiplication Coefficient for Cr Calculation These bits specify the G multiplication coefficient for the Cr data calculation equation in RGB → YCbCr color space conversion. (Initial value: -377) Specify a 13-bit signed integer obtained by multiplying the desired coefficient value by 2^ CRCLSFT. The MSB is a sign bit.	R/W
15:13	—	These bits are read as 0. The write value should be 0.	R/W
28:16	CRCLBP[12:0]	B Multiplication Coefficient for Cr Calculation These bits specify the B multiplication coefficient for the Cr data calculation equation in RGB → YCbCr color space conversion. (Initial value: -73) Specify a 13-bit signed integer obtained by multiplying the desired coefficient value by 2^ CRCLSFT. The MSB is a sign bit.	R/W
31:29	—	These bits are read as 0. The write value should be 0.	R/W

Note: S-TYPE-3, P-TYPE-3

68.2.25.9 CRCCR3 : RGB to Cr Calculation Setting Register 3

Base address: VIN0 = 0x4034_7400
VIN0_NS = 0x5034_7400

Offset address: 0x248

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	CRCLSFT[4:0]					CRCLHEN	—	—	—	—	—	—	—
Value after reset:	1	0	0	0	1	0	1	0	0	0	0	0	0	0	0	1

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	CRCLAP[11:0]											
Value after reset:	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
11:0	CRCLAP[11:0]	Cr Calculation Data Normalized Additional Value These bits set the Cr data addition constant for RGB → YCbCr color space conversion. (Initial value: 2048) Do not change from the initial value .	R/W
15:12	—	These bits are read as 0. The write value should be 0.	R/W
16	—	This bit is read as 1. The write value should be 1.	R/W
22:17	—	These bits are read as 0. The write value should be 0.	R/W
23	CRCLHEN	Cr Calculation Shift Down Result Round-Off Enable This bit enables round-off process for Cr data calculation in RGB → YCbCr color space conversion 0: Round down to down shift process 1: Round-off to down shift process is enabled.	R/W
28:24	CRCLSFT[4:0]	Cr Calculation Shift Down Volume These bits set the amount of down shift for Cr calculation in RGB → YCbCr color space conversion (Initial value: 10) Unit: Bit shift count Do not change from the initial value .	R/W
30:29	—	These bits are read as 0. The write value should be 0.	R/W
31	—	This bit is read as 1. The write value should be 1.	R/W

Note: S-TYPE-3, P-TYPE-3

68.3 Operation

68.3.1 Input Interface

The VIN captures video data in the MIPI CSI-2 interface and stores it in external memory. The following tables show the interface and data format supported.

Table 68.11 Supported Interfaces

Input interface	MIPI CSI-2	YCbCr-422	8 bits	Supported
			10 bits	Supported
		RGB-888	24 bits	Supported
		8-bit user defined data (RAW8)	8 bits	Supported
Output interface	RGB output	RGB-565	16 bits	Supported
		ARGB-1555	16 bits	Supported
		RGB-888	32 bits	Supported
		ARGB-8888	32 bits	Supported
	YCbCr output	YCbCr-422 multiplexed	8 bits	Supported
			10 bits	Supported
		YCbCr-422 separated Y/CbCr	Y: 8 bits CbCr: 8 bits	Supported
			Y: 10 bits CbCr: 8 bits	Supported
			Y: 10 bits CbCr: 10 bits	Supported
		YCbCr-422 separated only Y	8 bits	Supported
			10 bits	Supported
	RAW output	RAW8	8 bits	Supported

Note: 1. Dithering is performed for RGB-888 → RGB-565 or ARGB-1555 conversion.

2. RGB888->RGB-565 or ARGB-1555 conversion is done after 8-bit precision conversion and sent to memory.

3. For details of output interfaces, see section 68.3.7. Output Data Format.

68.3.2 Capture Mode

When the IM bit in the main control register (MC) is used to set the field for memory output target and then the continuous frame capture mode (CC) bit in the frame capture register (FC) is set to 1b, the captured data is sequentially transferred to the addresses set in MB1 to MB3 and the continuous frame capture mode is executed.

In this case, the latest data output MB* is indicated in the FBS bit of the module status register (MS).

When using continuous frame capture mode, the required register setting procedure is as follows

1. Set 1b to the module enable (ME) bit in the main control register (MC).
2. Set 1b to the continuous frame capture mode (CC) bit in the frame capture register (FC).

The sequence of start, stop, and resume of continuous frame capture mode operation causes the frame memory (MB*) to which data is output to transition from MB1 to MB2 to MB3 to MB1 without being initialized. See Figure 68.6.

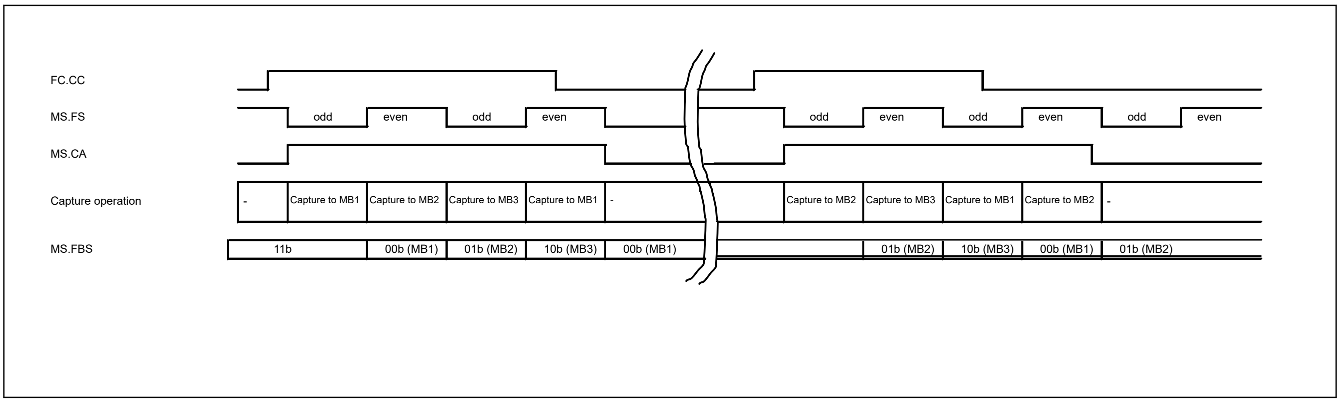


Figure 68.6 Data output frame memory in continuous frame capture mode (Example of operation)

68.3.3 Size Clipping

Image data that has been captured is pre-clipped according to the settings of the following registers:

start line pre-clip (SLPRC)

end line pre-clip (ELPRC)

start pixel pre-clip (SPPRC)

end pixel pre-clip (EPPRC).

For post-clipping after horizontal or vertical scaling, see section 68.2.22.4. UDS_CLIP_SIZE : UDS Output Size Clipping Registers.

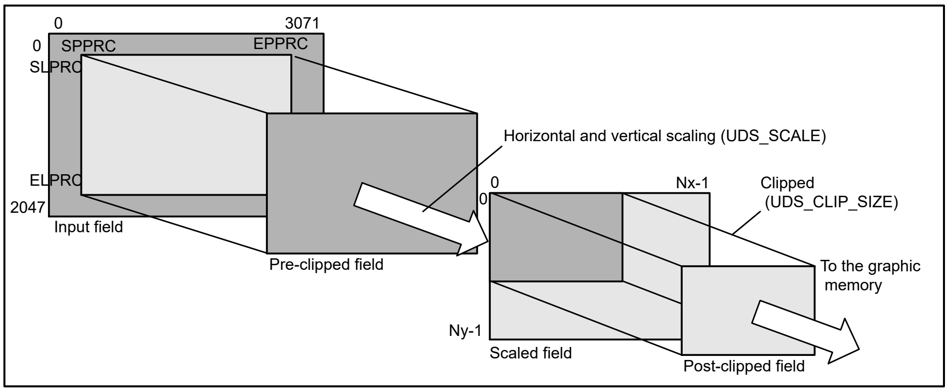


Figure 68.7 Example of clipping

For all the post-clipped lines, the lengths of individual lines written into the memory are defined by the image stride register (IS). The setting can be larger than the post-clipped frame width, but cannot be smaller than the width. The IS register must be filled with a value larger than the horizontal post-clipping width. The input field in the above figure shows an effective image area from the video decoder; the VIN does not allow anything exceeding the image area to be captured.

Note: Each of the following registers specifies a distance from the starting point in the effective image area: start line pre-clip (SLPRC), end line pre-clip (ELPRC), start pixel pre-clip (SPPRC), and end pixel pre-clip (EPPRC).

68.3.4 Scaling

(1) Vertical Scaling

For details on scaling in the vertical direction, see section 68.2.22.2. UDS_SCALE : Scaling Factor Registers.

(2) Horizontal Scaling

For details on scaling in the horizontal direction, see section 68.2.22.2. UDS_SCALE : Scaling Factor Registers.

(3) Scaling constraint

When scaling is performed, there are restrictions on the magnification rate depending on the input timing and the frequency of the aclk.

(4) Bandlimited

Bandwidth constraints vary depending on input timing, magnification factor, and frequency of the aclk. Please ensure the bandwidth as shown below.

1. Transmittable bandwidth of depth * (EPPRC[11:0] - SPPRC[11:0] + 1) * hscale bit for the period shown in Table 68.12.
2. To prevent the internal FIFO from overflowing, it is necessary to transmit at least α bit per aclk.
   α > (depth * (EPPRC[11:0] - SPPRC[11:0] + 1) * hscale - 32768 bit) / Send request period
   If α is negative, only the bandwidth in "1." should be reserved.

Table 68.12 Scaling bandwidth constraint

No vertical scaling: when vscale ≤ 1 and hscale ≤ (aclk / dotclk)		
1	Period (unit: aclk)	HTOTAL * aclk (MHz) / dotclk (MHz) cycle (1H)
2	Transmission request per aclk (unit bit)	depth * dotclk / aclk * hscale
3	Transmission request period (unit: aclk)	(EPPRC[11:0] − SPPRC[11:0] + 1)*aclk/dotclk
No vertical scaling: when vscale ≤ 1 and hscale > (aclk / dotclk)		
4	Period (unit: aclk)	HTOTAL * aclk (MHz) / dotclk (MHz) cycle (1H)
5	Transmission request per aclk (unit bit)	depth
6	Transmission request period (unit: aclk)	(EPPRC[11:0] − SPPRC[11:0] + 1) * hscale
With vertical scaling: when vscale > 1		
7	Period (unit: aclk)	(EPPRC[11:0] − SPPRC[11:0] + 1) * hscale + interval cycle
8	Transmission request per aclk (unit bit)	depth
9	Transmission request period (unit: aclk)	(EPPRC[11:0] − SPPRC[11:0] + 1) * hscale

depth: Bits per pixel at capture

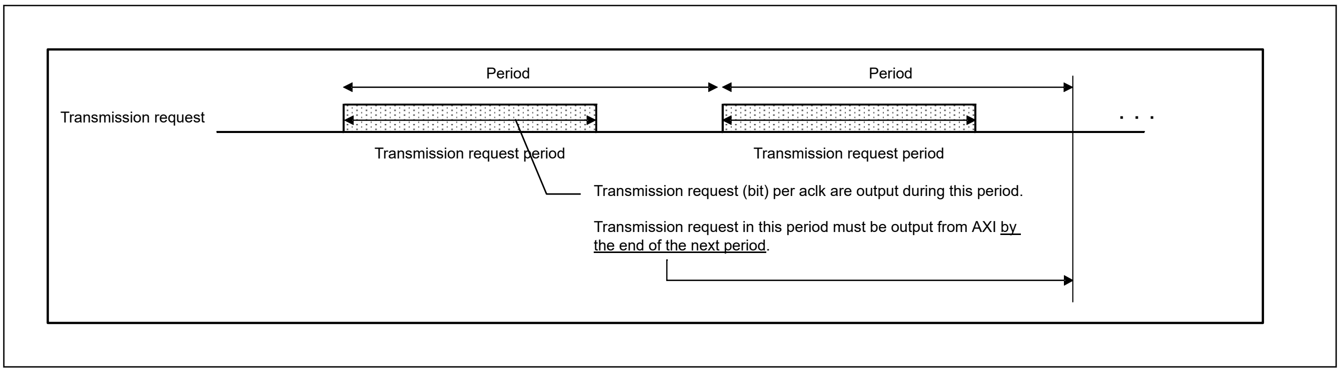


Figure 68.8 Transmission request when using scaler

68.3.5 Color Conversion Function

(1) YC-RGB Color Conversion

When the data input format is YCbCr, set the BPS bit in the MC register to 0 to convert YCbCr data to RGB data. It is carried out according to the matrix coefficients set in the CSCE1, CSCE2, CSCE3, and CSCE4 registers.

The color conversion input is extended to 12 bits by the data extension selection register (CSI_IFMD.DES0).

Here, if the BPS bit in MC is set to 1, the data is stored in memory as it is in YCbCr format.

R = CSCE1.YMUL2[13:0] × (Y- CSCE2.YSUB2 [11:0])

+ CSCE3.RCRMUL2 [13:0] × (Cr- CSCE2.CSUB2 [11:0])

G = CSCE1.YMUL2[13:0] × (Y- CSCE2.YSUB2 [11:0])

- CSCE3.GCRMUL2 [13:0] × (Cr- CSCE2.CSUB2 [11:0])

- CSCE4.GCBMUL2 [13:0] × (Cb- CSCE2.CSUB2 [11:0])

B = CSCE1.YMUL2[13:0] × (Y- CSCE2.YSUB2 [11:0])

+ CSCE4.BCBMUL2 [13:0] × (Cb- CSCE2.CSUB2 [11:0])

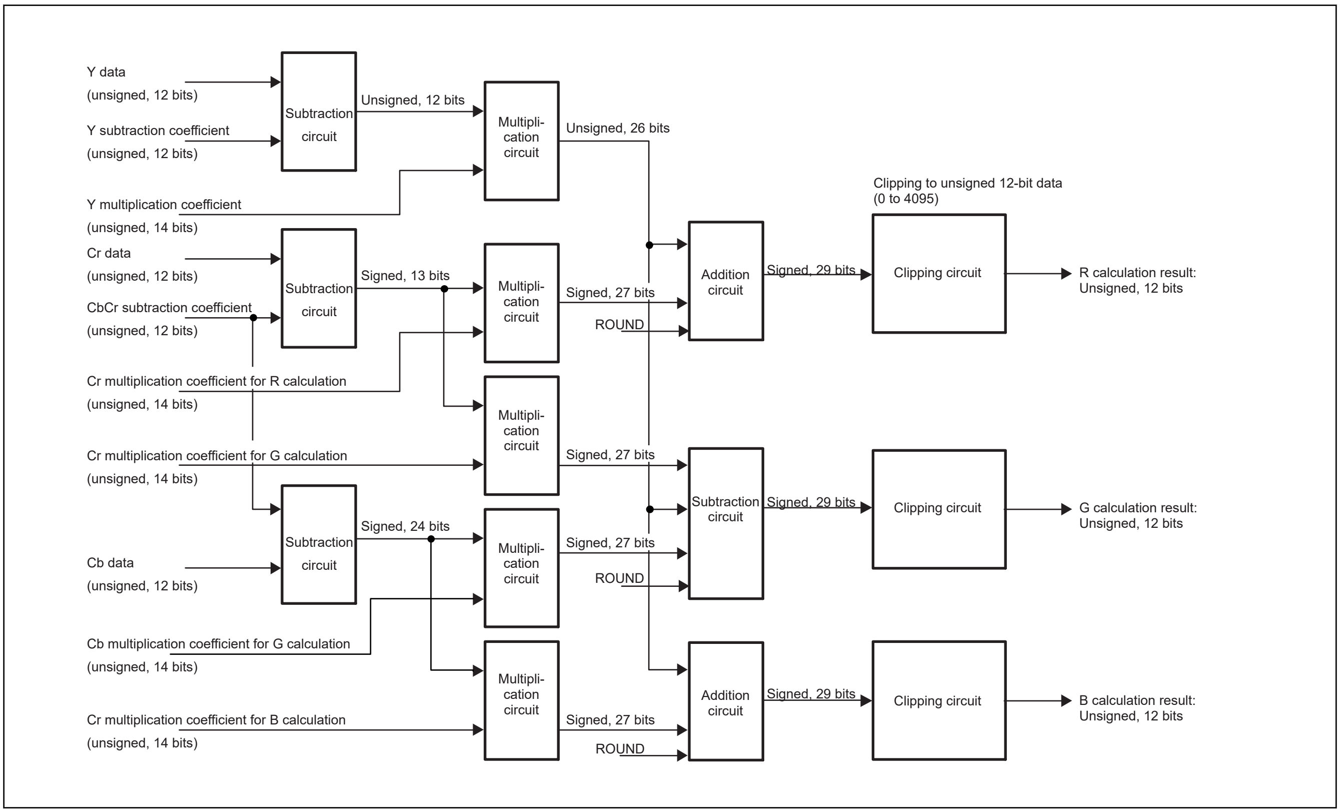


Figure 68.9 YCbCr to RGB calculation circuit configuration

Examples of color space that can be set in YC-RGB conversion coefficient registers are shown below. See the register descriptions for details of the coefficients.

In addition, the rounding process (CSCE1.ROUND) bit for the RGB calculation result allows the user to select between truncation and rounding.

Table 68.13 Examples of YC-RGB conversion coefficient register settings

YC-RGB conversion coefficient	YMUL2	YSUB2	CSUB2	RCRMUL 2	GCRMUL 2	GCBMUL 2	BCBMUL 2
ITU-R BT.601 (initial value) 16 ≤ Y ≤ 235, 16 ≤ Cb, Cr ≤ 240	1.164	256	2048	1.596	0.813	0.392	2.017
Luminance expansion example 1 ≤ Y ≤ 254, 16 ≤ Cb, Cr ≤ 240	1.008	16	2048	1.596	0.813	0.392	2.017

Specify an integer in each addition coefficient bit field. For each multiplication coefficient, specify a value obtained by multiplying the desired coefficient value by 4096.

Example: When the desired multiplication coefficient is 1.164

1.164 × 4096 = 4767 (set value: 0x129F)

Note: In order to process the data outside the range prescribed by the ITU-R BT.601 standard, set the CLP[1:0] bits in MC to 11.

(2) RGB-YC Color Conversion

When the data input format is RGB, set the BPS bit in MC to 0 for color conversion of RGB data into YCbCr data format. Color conversion from RGB to YCbCr is done according to the matrix coefficients set in RGB-YC conversion coefficient registers (YCCR1-YCCR3/CBCCR1-CBCCR3/CRCCR1-CRCCR3).

The color conversion input is extended to 12 bits by the data extension selection register (CSI_IFMD.DES0).

If 1 is set to the BPS bit in MC, the data will be stored in the memory as it is in RGB format.

Y = ((YCLRP × R + YCLGP × G + YCLBP × B) × 2 ^ YCLSFT) + YCALP

Cb = ((CBCLRP × R + CBCLGP × G + CBCLBP × B) × 2 ^ CBCLSFT) + CBCALP

Cr = ((CRCLRP × R + CRCLGP × G + CRCLBP × B) × 2 ^ CRCLSFT) + CRCALP

The following data rounding functions of RGB-YCbCr color conversion function can be independently set to each pixel. The circuit configuration of Y data is given below.

Table 68.14 Data rounding functions of RGB-YC color conversion function

Function	Symbol	Function
Multiplication result shift down amount	YCLSFT[4:0]	Amount of shift down in the matrix multiplication result
Rounding off enable	YCLHEN	Enables/disables rounding off to the shift down amount.

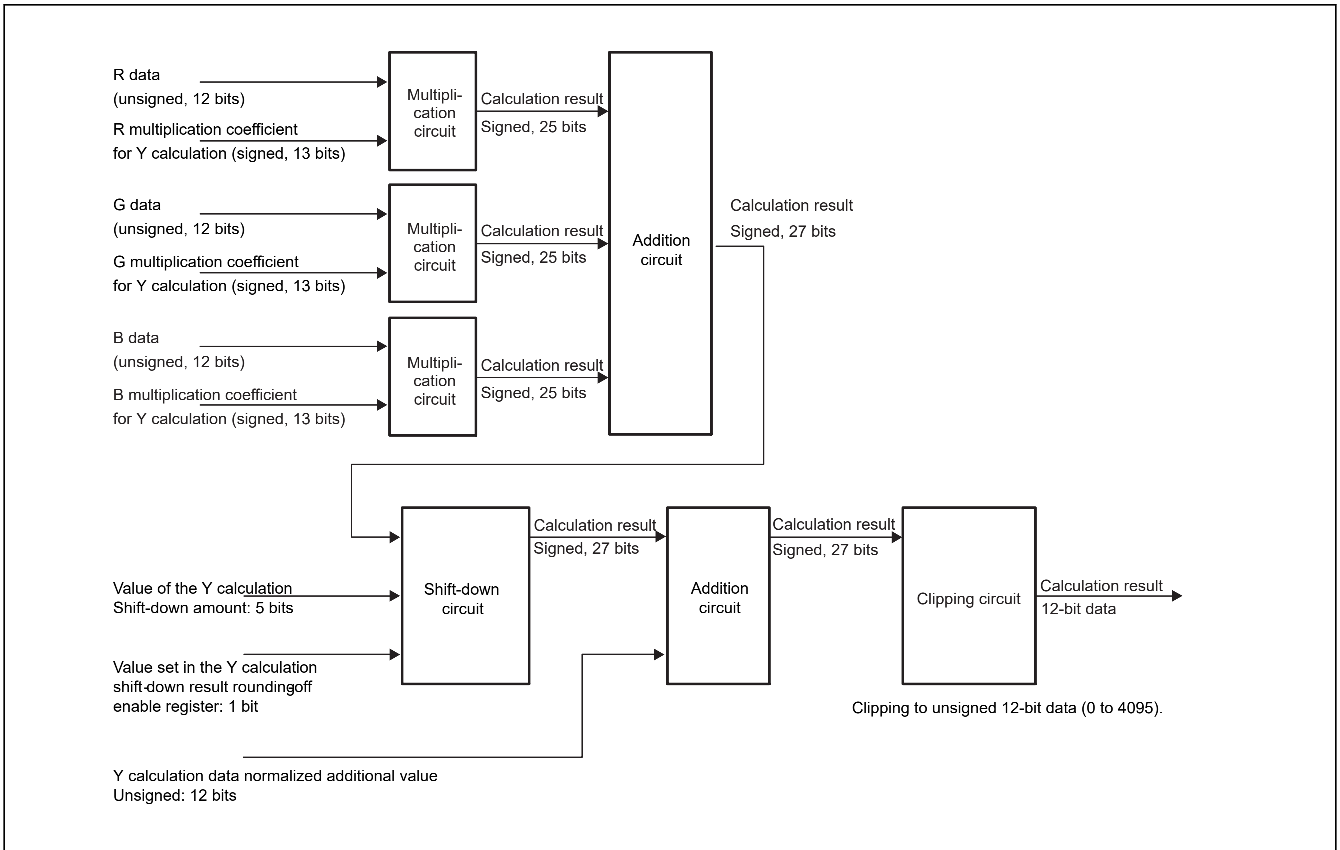


Figure 68.10 Y data circuit configuration of RGB to YCbCr color conversion function

68.3.6 Image Data Format Conversion Functions

(1) Lookup Table (LUT) Precision Conversion Function

Set the LUTE bit in MC to 1 to enable table conversion of each pixel data of Y, Cb, and Cr or R, G, and B data after color conversion.

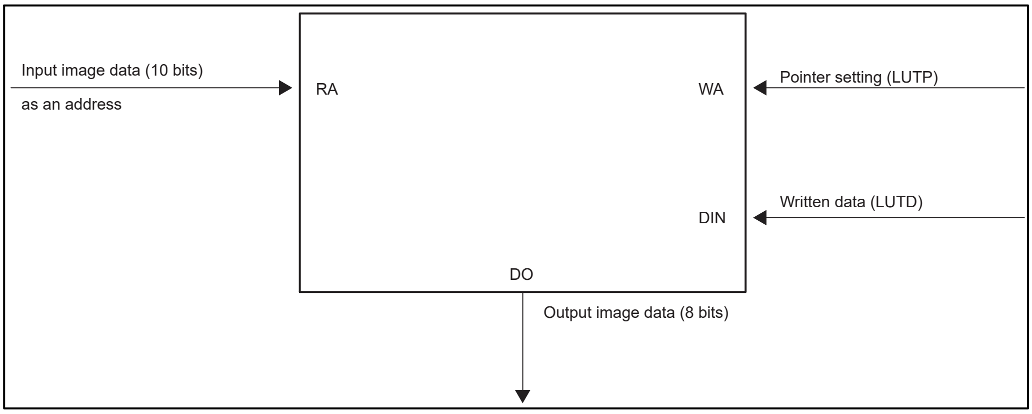


Figure 68.11 Lookup table

Pointer Access:

The pointer is incremented every time a write access to the LUTD register is made.

The incremented pointer value is read when the LUTP register is read. Since the pointer is not incremented at read access to the LUTD register, the pointer should be set just before reading.

The procedure given below must be followed to access the lookup table.

The lookup table cannot be accessed from the CPU during the conversion process.

(For write/read access to the LUT from the CPU, the LUTE bit in MC should be 0.)

Write access:

1. Write the access destination address in LUTP.
2. Write the data in LUTD. Thereby the data will be reflected in the lookup table.
   Also, the pointer will automatically increase by 1.
3. Writing to LUTD will consecutively update the lookup table.

Read Access:

1. Write the access destination address in LUTP.
2. Read LUTD. This reading allows the address data written in LUTP to be acquired.

[Notes on Using the Lookup Table]

1. Set the lookup table only after capture operation has stopped.
2. Write all before using the lookup table.
3. Data set to the lookup table should be within the range compatible to the input bit width.

(2) YCbCr422 to YC Separation Function

When transferring YCbCr data to memory, Y data and CbCr data can be separated and transferred to different address spaces.

To perform YCbCr separation transfer, set the DTMD[1:0] bits in the DMR register to 10b. In this case, the Y data will be transferred to the address set in the memory base address register and CbCr data will be transferred to the address obtained by adding the value set in the UVAOF register to the memory base address register.

If the YMODE[0] bits in the DMR register are set, only Y data will be transferred to memory and CbCr data is not transferred to memory.

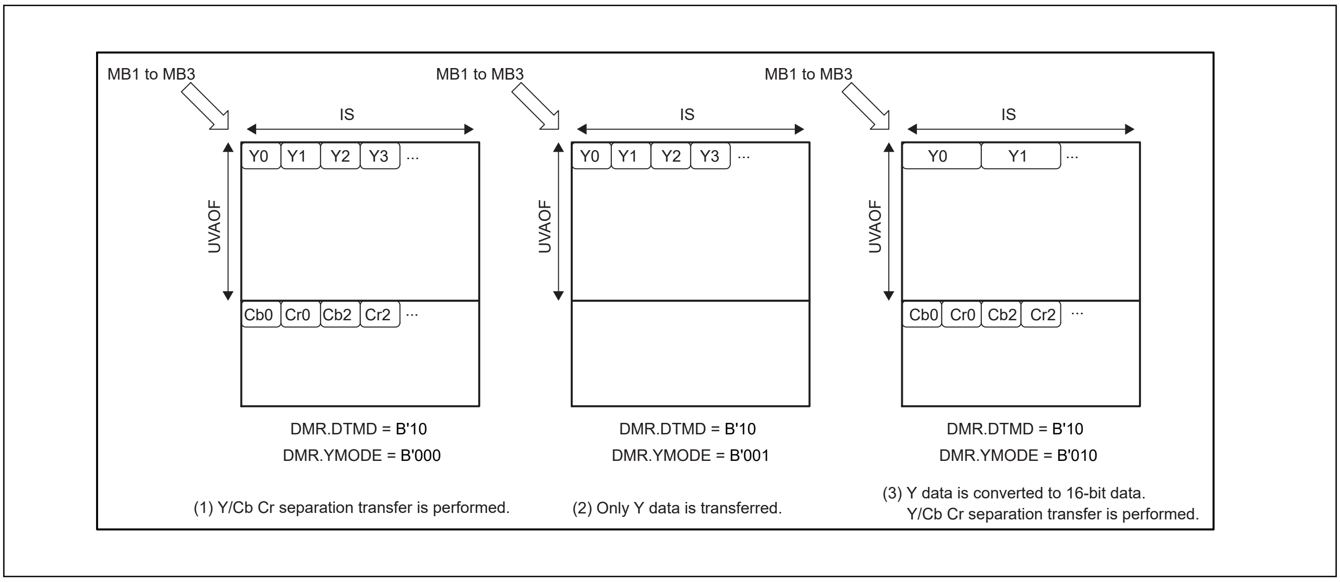


Figure 68.12 Y/Cb/Cr separation in big endian

(3) Dithering Function

Dithering is performed when the internal RGB-888 format after color space conversion is converted to the RGB-565 or ARGB-1555 format. The dithering mode can be selected with the DC[1:0] bits in MC.

(a) Dithering with Cumulative Addition

Set the DC[1:0] bits in MC to 00b to perform dithering using the cumulative addition method in horizontal pixel units.

Dithering to RGB565	Dithering to ARGB1555
Rn[7:3] ≤ (Rx[7:0] + Rn-1[2:0]) >> 3 Gn[7:2] ≤ (Gx[7:0] + Gn-1[1:0]) >> 2 Bn[7:3] ≤ (Bx[7:0] + Bn-1[2:0]) >> 3 (Rn, Gn, Bn) = Output RGB565 pixel (Rx, Gx, Bx) = Input RGB888 pixel (Rn-1, Gn-1, Bn-1) = Pseudo random error (LSB of the cumulative error)	Rn[7:3] ≤ (Rx[7:0] + Rn-1[2:0]) >> 3 Gn[7:3] ≤ (Gx[7:0] + Gn-1[2:0]) >> 3 Bn[7:3] ≤ (Bx[7:0] + Bn-1[2:0]) >> 3 (Rn, Gn, Bn) = Output RGB555 pixel (Rx, Gx, Bx) = Input RGB888 pixel (Rn-1, Gn-1, Bn-1) = Pseudo random error (LSB of the cumulative error)

If overflow occurs due to dither addition, it is clipped to all1 and output.

(b) Ordered Dithering 1

Set the DC[1:0] bits in MC to 01b to perform dithering using the ordered dithering method.

Dithering to RGB565	Dithering to ARGB1555
Rn[7:3] ≤ (Rx[7:0] + $D42_{yx}$) >> 3 Gn[7:2] ≤ (Gx[7:0] + $D22_{yx}$) >> 2 Bn[7:3] ≤ (Bx[7:0] + $D42_{yx}$) >> 3 (Rn, Gn, Bn) = Output RGB565 pixel (Rx, Gx, Bx) = Input RGB888 pixel	Rn[7:3] ≤ (Rx[7:0] + $D42_{yx}$) >> 3 Gn[7:3] ≤ (Gx[7:1] + $D22_{yx}$) >> 2 Bn[7:3] ≤ (Bx[7:0] + $D42_{yx}$) >> 3 (Rn, Gn, Bn) = Output RGB555 pixel (Rx, Gx, Bx) = Input RGB888 pixel

If overflow occurs due to dither addition, it is clipped to all1 and output.

$D22_{yx}$, $D42_{yx}$ indicate the following dither patterns

$$D22 = \begin{bmatrix} 03 \\ 21 \end{bmatrix}, D42 = \begin{bmatrix} 0347 \\ 2165 \end{bmatrix}$$

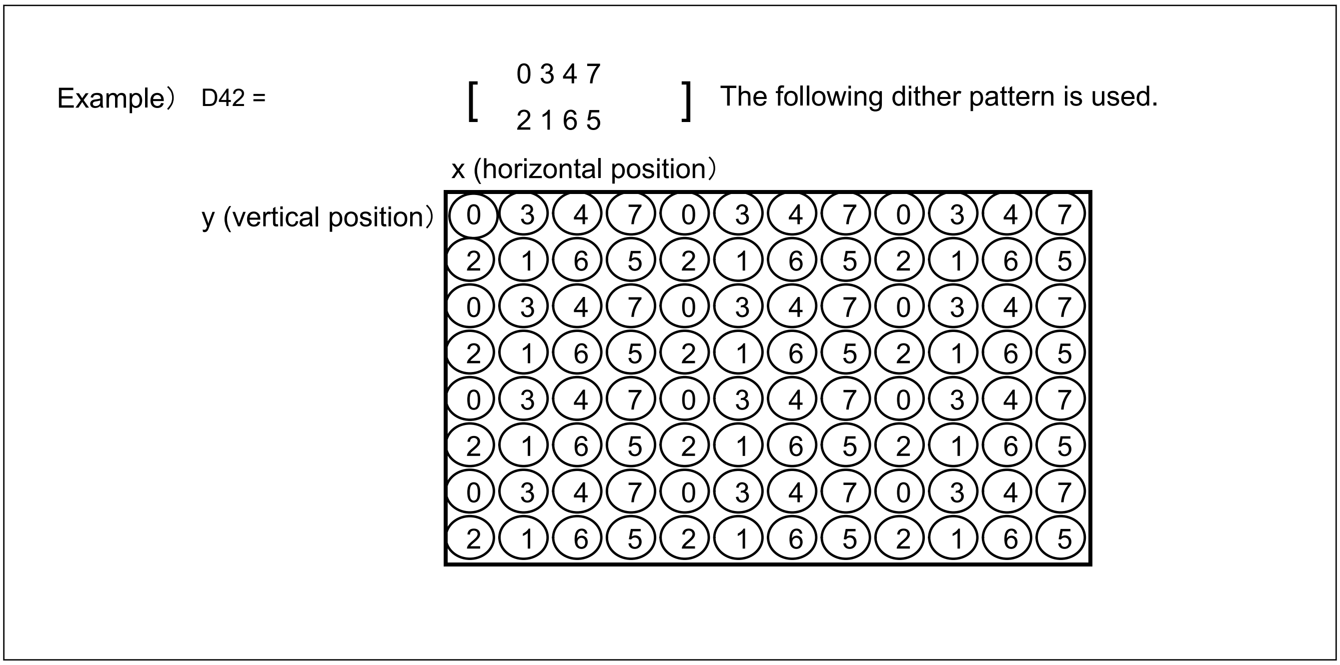


Figure 68.13 Example D42 dither pattern

(c) Ordered Dithering 2

Set the DC[1:0] bits in MC to 11b to perform dithering using the ordered dithering method.

Dithering to RGB565	Dithering to ARGB1555
Rn[7:3] ≤ (Rx[7:1] + D42-f $_{yx}$) >> 2 Gn[7:2] ≤ (Gx[7:0] + D42-f $_{yx}$) >> 2 Bn[7:3] ≤ (Bx[7:1] + D42-f $_{yx}$) >> 2 (Rn, Gn, Bn) = Output RGB565 pixel (Rx, Gx, Bx) = Input RGB888 pixel	Rn[7:3] ≤ (Rx[7:1] + D42-f $_{yx}$) >> 2 Gn[7:3] ≤ (Gx[7:1] + D42-f $_{yx}$) >> 2 Bn[7:3] ≤ (Bx[7:1] + D42-f $_{yx}$) >> 2 (Rn, Gn, Bn) = Output RGB555 pixel (Rx, Gx, Bx) = Input RGB888 pixel

If overflow occurs due to dither addition, it is clipped to all1 and output.

When MC.IM is 2'b01, f transitions 0 → 1 → 2 → 3 → 0 for each field.

When MC.IM is not 2'b01, f transitions 0 → 1 → 2 → 3 → 0 for each odd field.

D42-f $_{yx}$, indicates the following dither pattern.

If DC2 bit is set to 1b when DC[1:0] of MC is set to 11b, dithering is always performed at D42-0 yx.

$$D42-0=\begin{bmatrix}3021\\2130\end{bmatrix}$$

$$D42-1=\begin{bmatrix}0213\\1302\end{bmatrix}$$

$$D42-2=\begin{bmatrix}2130\\3021\end{bmatrix}$$

$$D42-3=\begin{bmatrix}1302\\0213\end{bmatrix}$$

68.3.7 Output Data Format

The VIN can output image data in the following formats. The figures in this section assume that data is stored in unified memory in little endian.

(1) YC: YCbCr-422, 8 bits

The 8-bit YUV image data in the YC (YCbCr) = 4:2:2 format is shown below. YC data can be switched between the UYVY format and YUYV format through the BPSM bit in DMR.

- BPSM = 0 in DMR: UYVY format

8-bit YCbCr-422 data (UYVY format)																
D63 to D48	63							56	55							48
Image data 3 and 4	Y3[7:0]								Cr2[7:0]							
D47 to D32	47							40	39							32
Image data 3 and 4	Y2[7:0]								Cb2[7:0]							
D31 to D16	31							24	23							16
Image data 1 and 2	Y1[7:0]								Cr0[7:0]							
D15 to D0	15							8	7							0
Image data 1 and 2	Y0[7:0]								Cb0[7:0]							

	DMR.YMODE[2:0]	DMR.EXRGB	DMR.YC_THR	DMR.DTMD[1:0]	DMR.BPSM
YCbCr422 (8 bit) transfer	000b	0	0	00b	0

- BPSM = 1 in DMR: YUYV format

8-bit YCbCr-422 data (YUYV format)																
D63 to D48	63							56	55							48
Image data 3 and 4	Cr2[7:0]								Y3[7:0]							
D47 to D32	47							40	39							32
Image data 3 and 4	Cb2[7:0]								Y2[7:0]							
D31 to D16	31							24	23							16
Image data 1 and 2	Cr0[7:0]								Y1[7:0]							
D15 to D0	15							8	7							0
Image data 1 and 2	Cb0[7:0]								Y0[7:0]							

	DMR.YMODE[2:0]	DMR.EXRGB	DMR.YC_THR	DMR.DTMD[1:0]	DMR.BPSM
YCbCr422 (8 bit) transfer	000b	0	0	00b	1

(2) YC: YCbCr-422, 10 bits

The 10-bit YUV image data in the YC (YCbCr) = 4:2:2 format is shown below.

- 10-bit YCbCr-422 data: (UYVY format)

10-bit YCbCr-422 data: (UYVY format)																
D127 to D112	127					122	121									112
Image data 8	0	0	0	0	0	0	Y3[9:0]									
D111 to D96	111					106	105									96
Image data 7	0	0	0	0	0	0	Cr2[9:0]									
D95 to D80	95					90	89									80
Image data 6	0	0	0	0	0	0	Y2[9:0]									

10-bit YCbCr-422 data: (UYVY format)																
D79 to D64	79					74	73									64
Image data 5	0	0	0	0	0	0	Cb2[9:0]									
D63 to D48	63					58	57									48
Image data 4	0	0	0	0	0	0	Y1[9:0]									
D47 to D32	47					42	41									32
Image data 3	0	0	0	0	0	0	Cr0[9:0]									
D31 to D16	31					26	25									16
Image data 2	0	0	0	0	0	0	Y0[9:0]									
D15 to D0	15					10	9									0
Image data 1	0	0	0	0	0	0	Cb0[9:0]									

	DMR.YMODE[2:0]	DMR.EXRGB	DMR.YC_THR	DMR.DTMD[1:0]
YCbCr422 (10 bit) transfer	000b	0	1	00b

(3) YC: YC Separation YCbCr-422, Y (8 bits)/C (8 bits)

This is 8-bit YUV image data in the YC separated YC (YCbCr) = 4:2:2 format.

If the YMODE[2:0] bits in the data mode register (DMR) are set to 000b or 001b, setting the DTMD[1:0] bits in DMR to 10b changes the format to 4:2:2. UV data is transferred to the address obtained by adding the value set in the UV address offset register (UVAOF) to the memory base address register. If the YMODE[2:0] bits in DMR are set to 001b, only Y data can be transferred and UV data cannot be transferred.

Y data																
D31 to D16	31							24	23							16
Image data 3 and 4	Y3[7:0]								Y2[7:0]							
D15 to D0	15							8	7							0
Image data 1 and 2	Y1[7:0]								Y0[7:0]							
Cb, Cr data																
D31 to D16	31							24	23							16
Image data 3 and 4	Cr2[7:0]								Cb2[7:0]							
D15 to D0	15							8	7							0
Image data 1 and 2	Cr0[7:0]								Cb0[7:0]							

	DMR.YMODE[2:0]	DMR.EXRGB	DMR.YC_THR	DMR.DTMD[1:0]	remarks
Y (8 bit)/CbCr (8 bit) separation transfer	000b	0	0	10b	The destination of CbCr is determined by the setting of the UVAOF register.
Transfer Y (8 bit) data only	001b	0	0	10b	

(4) YC: YC Separation YCbCr-422 Y (10 bits)/C (10 bits)

This is 10-bit YUV image data in the YC separated YC (YCbCr) = 4:2:2 format.

If the YMODE[2:0] bits in the data mode register (DMR) are set to 000b, setting the DTMD[1:0] bits in DMR to 10b changes the format to 4:2:2. UV data is transferred to the address obtained by adding the value set in the UV address offset register (UVAOF) to the memory base address register.

Y data (10 bits)																
D63 to D48	63					58	57									48
Image data 4	0	0	0	0	0	0	Y3[9:0]									
D47 to D32	47					42	41									32
Image data 3	0	0	0	0	0	0	Y2[9:0]									
D31 to D16	31					26	25									16
Image data 2	0	0	0	0	0	0	Y1[9:0]									
D15 to D0	15					10	9									0
Image data 1	0	0	0	0	0	0	Y0[9:0]									
Cb, Cr data (10 bits)																
D63 to D48	63					58	57									48
Image data 4	0	0	0	0	0	0	Cr2[9:0]									
D47 to D32	47					42	41									32
Image data 3	0	0	0	0	0	0	Cb2[9:0]									
D31 to D16	31					26	25									16
Image data 2	0	0	0	0	0	0	Cr0[9:0]									
D15 to D0	15					10	9									0
Image data 1	0	0	0	0	0	0	Cb0[9:0]									

	DMR.YMODE[2:0]	DMR.EXRGB	DMR.YC_THR	DMR.DTMD[1:0]	remarks
Y (10 bit)/ CbCr (10 bit) separation transfer	000b	0	1	10b	The destination of CbCr is determined by the setting of the UVAOF register.

(5) YC: YC Separation YCbCr-422, Y (10 bits)/C (8 bits)

This is 10-bit YUV image data in the YC separated YC (YCbCr) = 4:2:2 format.

If the setting of the YMODE[2:0] bits in the data mode register (DMR) is 010b or 011b, setting the DTMD[1:0] bits in DMR to 10b leads to the conversion of 10-bit CrCb data to 8-bit CrCb data and changing the format to 4:2:2.

UV data is transferred to the address obtained by adding the value set in the UV address offset register (UVAOF) to the memory base address register. If the YMODE[2:0] bits in DMR are set to 011b, only Y data can be transferred and UV data cannot be transferred.

Y data																
D63 to D48	63					58	57									48
Image data 4	0	0	0	0	0	0	Y3[9:0]									
D47 to D32	47					42	41									32
Image data 3	0	0	0	0	0	0	Y2[9:0]									
D31 to D16	31					26	25									16
Image data 2	0	0	0	0	0	0	Y1[9:0]									

D15 to D0	15					10	9									0
Image data 1	0	0	0	0	0	0	Y0[9:0]									
Cb, Cr data																
D31 to D16	31							24	23							16
Image data 3 and 4	Cr2[7:0]								Cb2[7:0]							
D15 to D0	15							8	7							0
Image data 1 and 2	Cr0[7:0]								Cb0[7:0]							

	DMR.YMODE[2:0]	DMR.EXRGB	DMR.YC_THR	DMR.DTMD[1:0]	remarks
Y (10 bit)/CbCr (8 bit) separation transfer	010b	0	0	10b	The destination of CbCr is determined by the setting of the UVAOF register.
Transfer Y (10 bit) data only	011b	0	0	10b	

(6) 16 Bits/Pixel: RGB-565

The RGB levels are expressed through 5 bits for R, 6 bits for G, and 5 bits for B.

16-bit/pixel data (RGB data) format																
D15 to D0	15				11	10					5	4				0
Image data	R[4:0]					G[5:0]						B[4:0]				

	DMR.YMODE[2:0]	DMR.EXRGB	DMR.YC_THR	DMR.DTMD[1:0]	remarks
RGB565 (16 bit/pixel) format	000b	0	0	00b	

(7) 16 Bits/Pixel: ARGB-1555

The ARGB levels are expressed through 1 bit for A, 5 bits for R, 5 bits for G, and 5 bits for B.

Set the DTMD[1:0] bits in the data mode register (DMR) to 01b to specify conversion to ARGB-1555, and specify the A value in the ABIT bit in DMR.

16-bit/pixel data (ARGB data) format																
D15 to D0	15	14				10	9				5	4				0
Image data	A	R[4:0]					G[4:0]					B[4:0]				

	DMR.YMODE[2:0]	DMR.EXRGB	DMR.YC_THR	DMR.DTMD[1:0]	remarks
ARGB1555 (16 bit/pixel) format	000b	0	0	01b	The alpha bit is set by the ABIT bit in DMR.

(8) 32 Bits/Pixel: RGB-888

The RGB levels are expressed through 8 bits for R, 8 bits for G, and 8 bits for B. Bits 31 to 24 are fixed to 0.

32-bit/pixel data (RGB data) format																
D31 to D16	31							24	23							16
Image data	0	0	0	0	0	0	0	0	R[7:0]							
D15 to D0	15							8	7							0

32-bit/pixel data (RGB data) format		
Image data	G[7:0]	B[7:0]

	DMR.YMODE[2:0]	DMR.EXRGB	DMR.YC_THR	DMR.DTMD[1:0]	remarks
RGB888 (32 bit/ pixel) format	000b	1	0	00b	

(9) 32 Bits/Pixel: ARGB-8888

The ARGB levels are expressed through 8 bits for A, 8 bits for R, 8 bits for G, and 8 bits for B. The A value specified by the A8BIT[7:0] bits in DMR is set in bits 31 to 24.

32-bit/pixel data (ARGB data) format																
D31 to D16	31							24	23							16
Image data	A[7:0]								R[7:0]							
D15 to D0	15							8	7							0
Image data	G[7:0]								B[7:0]							

	DMR.YMODE[2:0]	DMR.EXRGB	DMR.YC_THR	DMR.DTMD[1:0]	remarks
ARGB8888 (32 bit/ pixel) format	000b	1	0	01b	The alpha bit is set by the A8BIT bit in DMR.

(10) Raw: 8 bits

The format of 8-bit raw image data is shown below.

8-bit raw data format																
D63 to D48	63							56	55							48
Image data 7 and 8	P7[7:0]								P6[7:0]							
D47 to D32	47							40	39							32
Image data 5 and 6	P5[7:0]								P4[7:0]							
D31 to D16	31							24	23							16
Image data 3 and 4	P3[7:0]								P2[7:0]							
D15 to D0	15							8	7							0
Image data 1 and 2	P1[7:0]								P0[7:0]							

	DMR.YMODE[2:0]	DMR.EXRGB	DMR.YC_THR	DMR.DTMD[1:0]	remarks
RAW8	000b	0	0	00b	The data is transferred in 4-pixel units.

68.3.8 Endianess Conversion

The VIN stores captured data in memory in little endian with the initial settings. Set the EN bit in the main control register (MC) to 1 to convert data into big endian before storing in memory.

Endian conversion is performed in 1-byte, 2-byte, and 4-byte units, as shown in Figure 68.14, according to the endian conversion units in Table 68.15.

For endian conversion of RAW, UYVY format, and YUYV format, data swap must be performed after endian conversion using MC.EN = 1b.

Figure 68.15 shows the relationship between MC.EN and DMR BPSM using the 8-bit UYVY format as an example.

The BPSM bit in the data mode register (DMR) is swapped in bytes.

Table 68.15 Endian conversion unit

Data output format	DMR.BPSM	Endian conversion unit
RAW8	1	2-byte unit
RGB565/ ARGB1555	—*1	2-byte unit
RGB888/ ARGB8888	—*1	4-byte unit
YCbCr422 8 bit UYVY	1	2-byte unit
YCbCr422 Y (8 bit)/C (8 bit)	—*1	1-byte unit
YCbCr422 Y (10 bit)/C (10 bit)	—*1	2-byte unit
YCbCr422 Y (10 bit)/C (8 bit)	—*1	Y: 2-byte unit C: 1-byte unit
Y8bit	—*1	1-byte unit
Y10bit	—*1	2-byte unit

Note 1. DMR.BPSM setting is invalid.

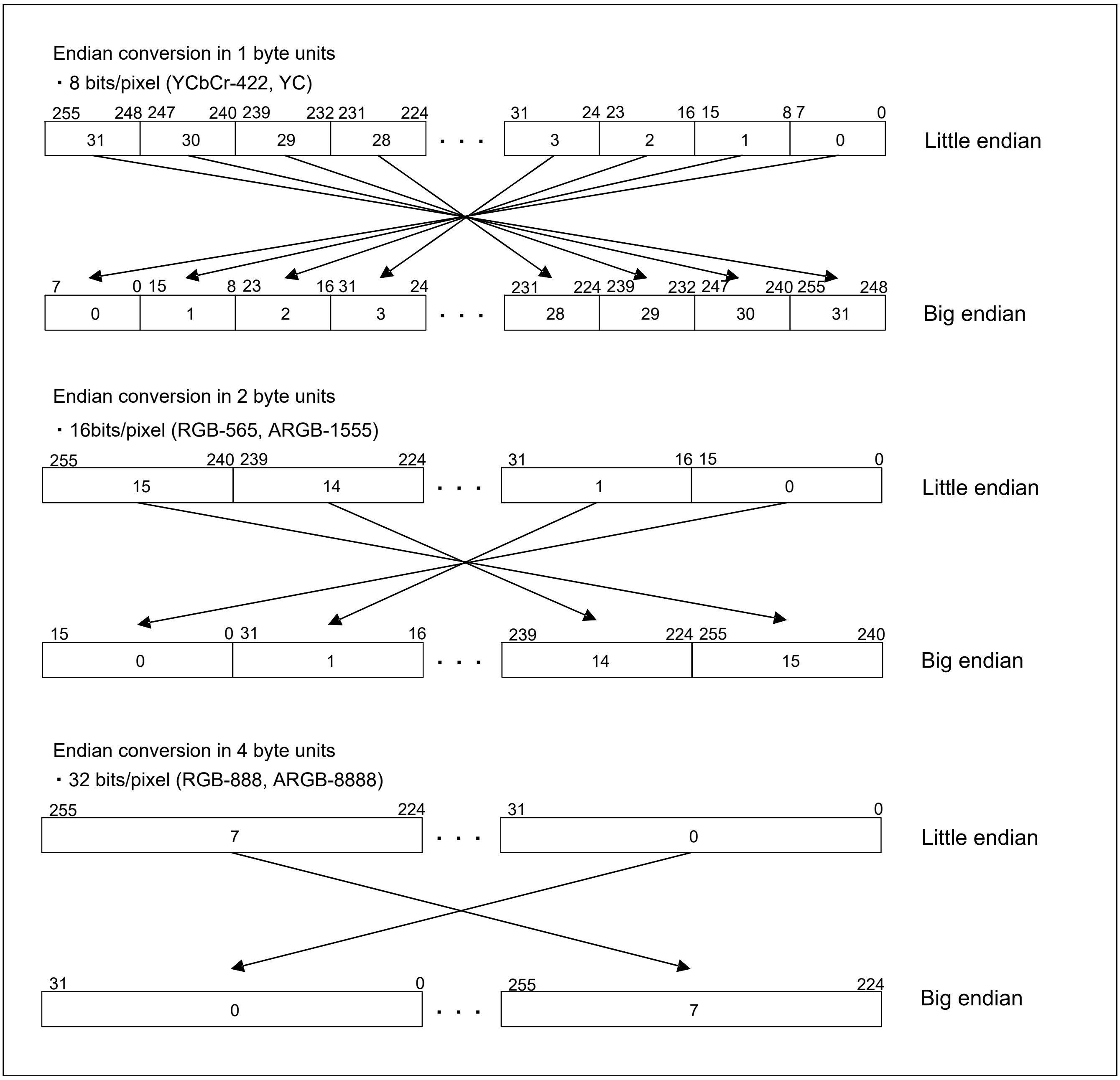


Figure 68.14 Data alignment conversion from little-endian to big-endian

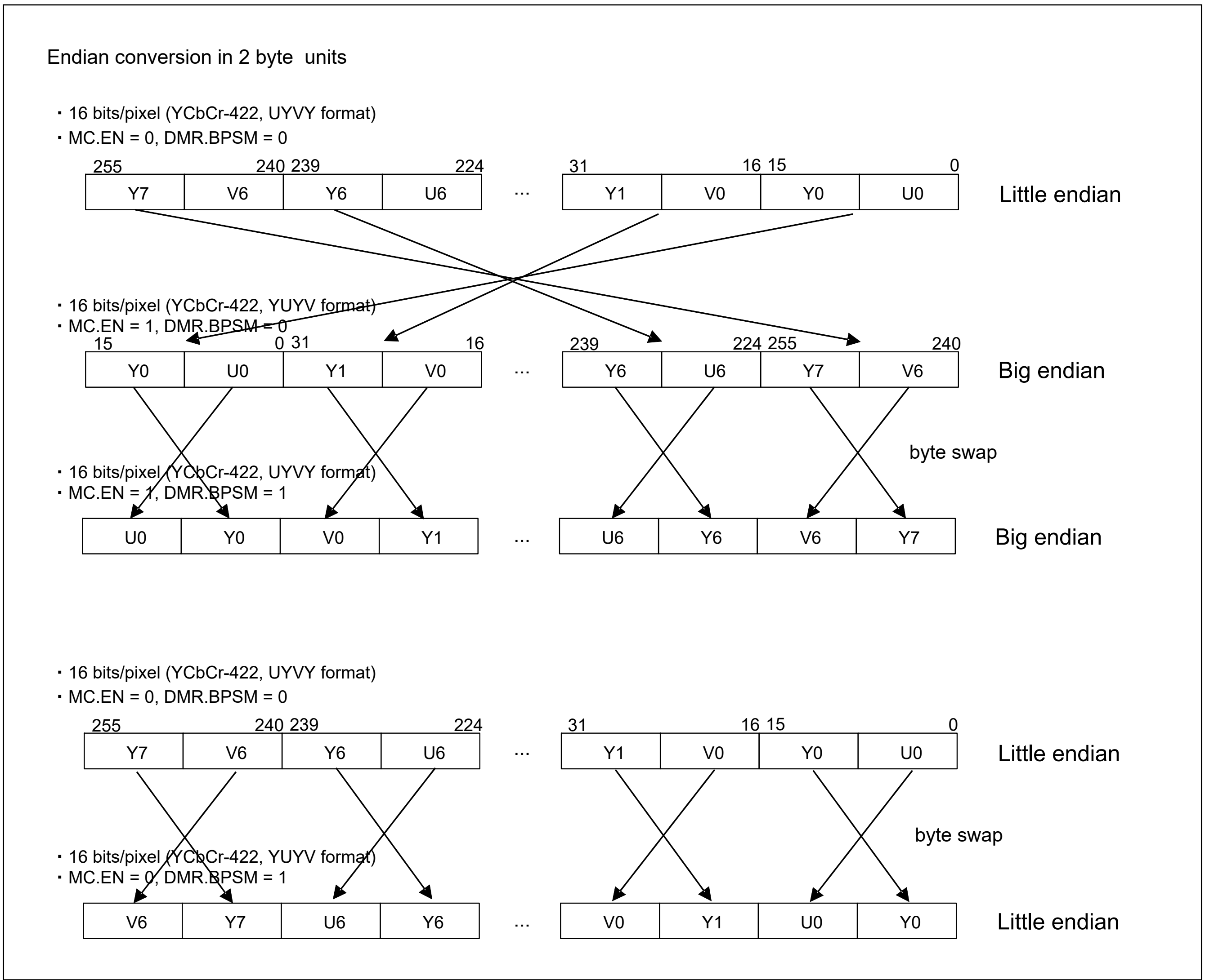


Figure 68.15 Endian conversion of 8-bit UYVY, YUYV format

68.3.9 Interrupts

VIN interrupts are shown below.

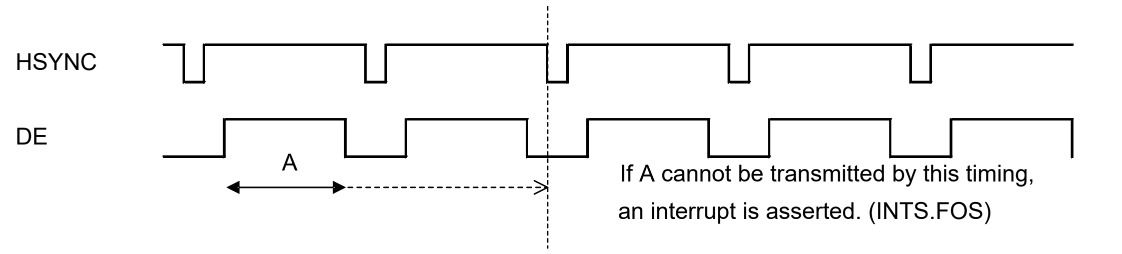
Table 68.16 Interrupt assignment

Event Name	Cause of interrupt
VIN0_IRQ	FIS2 or VFS or VRS or ARES or ROS or PRCLIPVES or PRCLIPHES or FMS, or FIS or SIS or EFS or FOS bit in INTS

68.3.10 Overflow

FIFO overflow (INTS.FOS) is asserted under the following conditions, so please reserve AXI bandwidth to avoid assertion.

1. When FIFO for AXI transfer overflows
2. If the transfer of captured data in the last line of one frame is not completed before the next FS is received
3. If the completion of data output for 1H period could not be transmitted before the start of the line after next. See Figure 68.16.
4. When scaling, follow the bandwidth constraints in section 68.3.4. Scaling for transmission. Failure to adhere may cause INTS.FOS to assert.



Note: For CSI-2, internal HSYNC is driven high when a long packet is received, and HSYNC is driven low when the long packet is received.

Figure 68.16 Overflow

68.3.11 Initialization Procedure

Figure 68.17 shows initialization procedure.

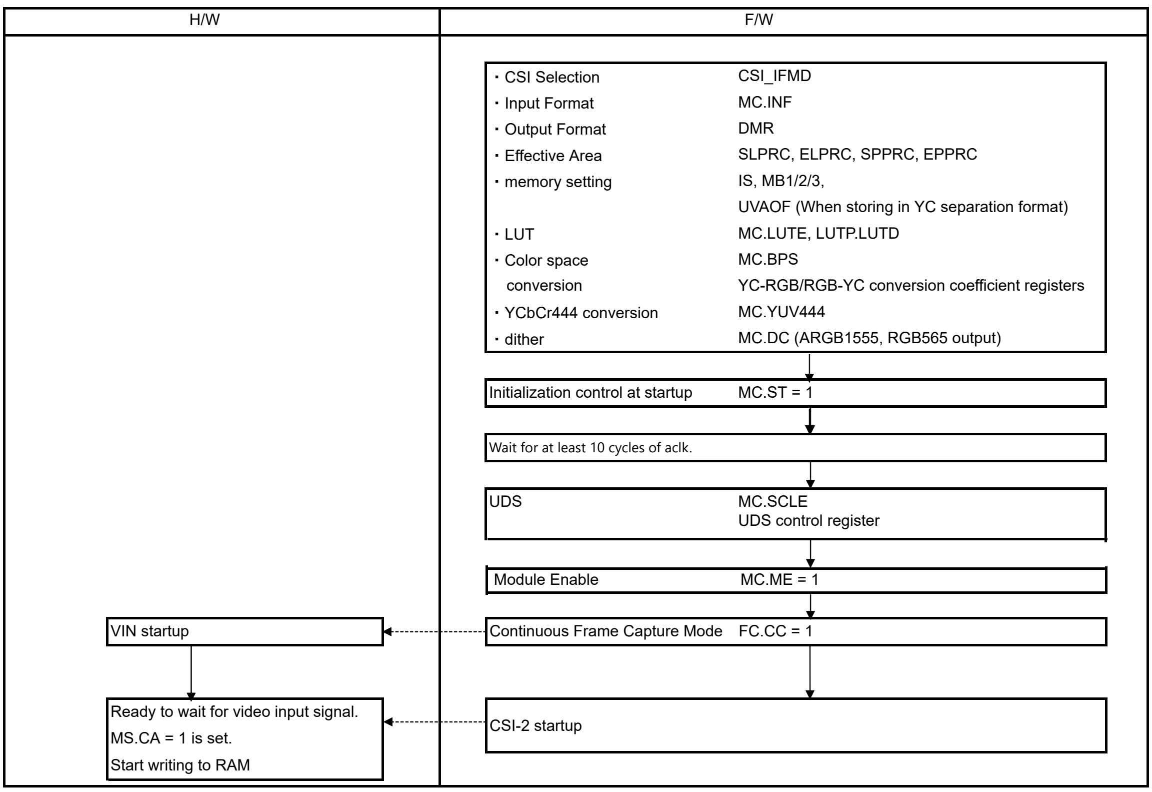


Figure 68.17 VIN initialization procedure (CC = 1 continuous frame capture mode)

68.3.12 Capture Stop Procedure

Figure 68.18 shows capture Stop Procedure.

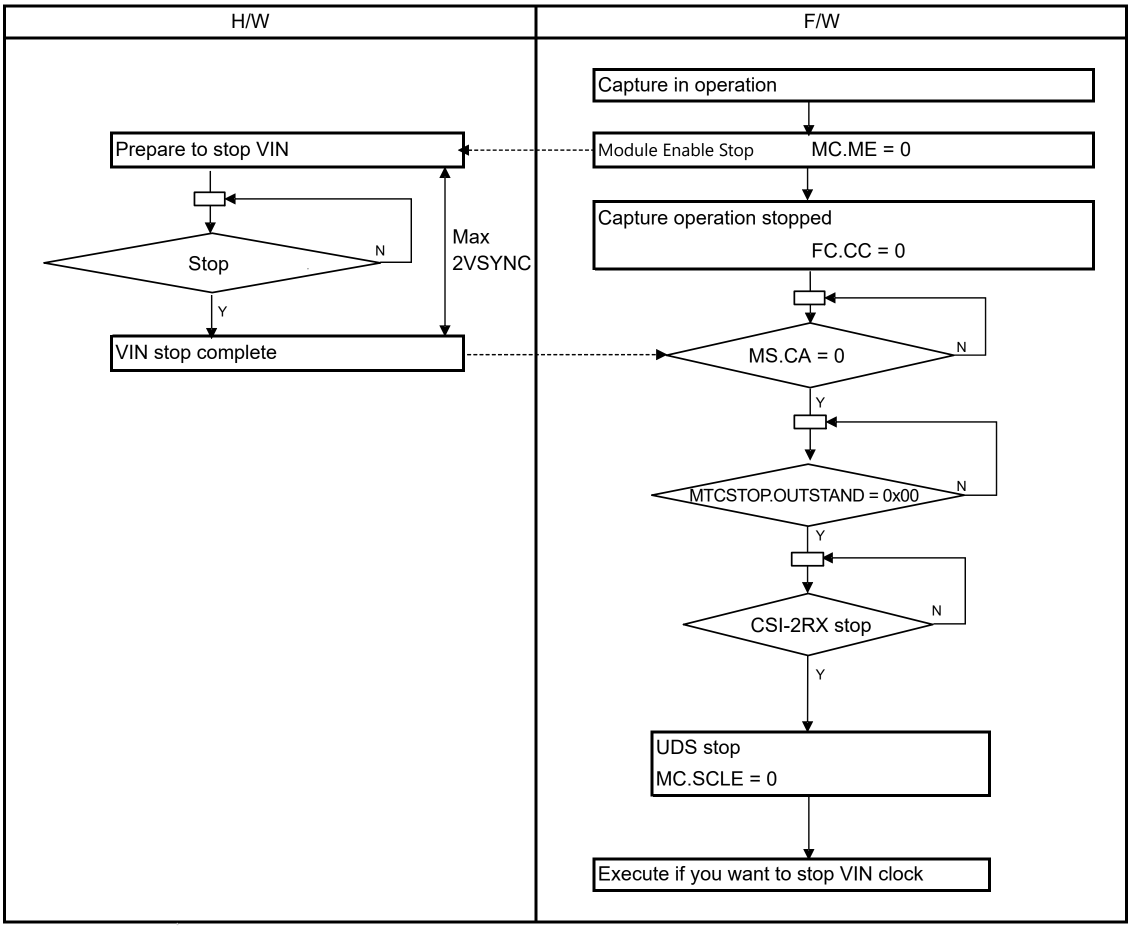


Figure 68.18 Stopping capture operation

68.3.13 Capture Restarting Procedure

Figure 68.19 shows capture Restarting Procedure.

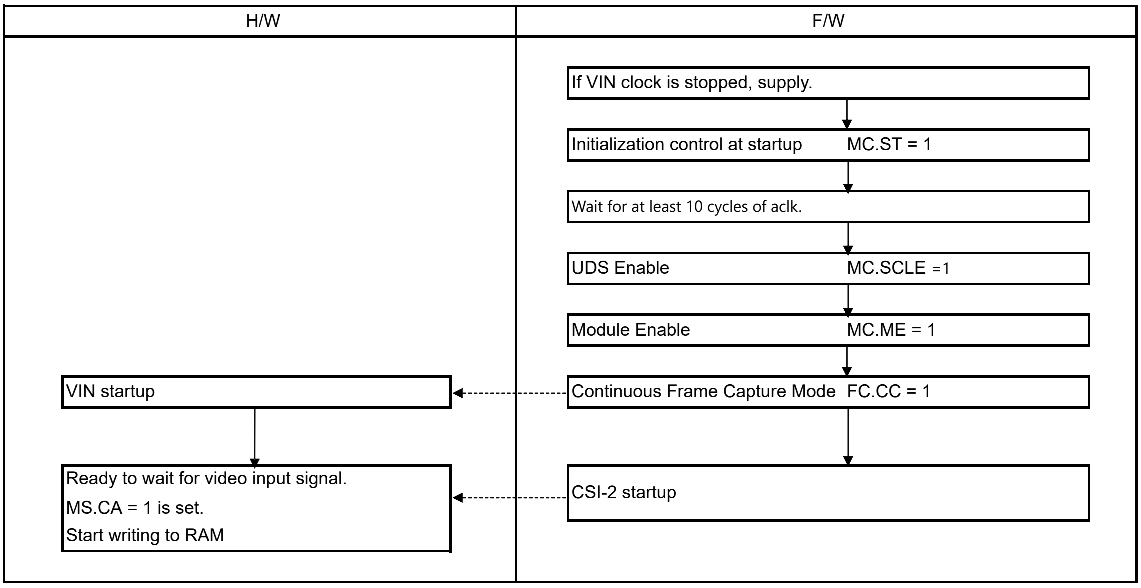


Figure 68.19 Capture restarting procedure

68.3.14 CSI control

In VIN, the data input from CSI-2 is selected by DT[5:0] and VC_SEL[3:0] bits in CSI_IFMD.

68.3.14.1 VSYNC Generation VSYNC

When a frame start short packet is received, VSYNC is driven high.

When a frame end short packet is received, VSYNC is driven low.

68.3.14.2 HSYNC Generation HSYNC

During the frame start short packet to frame end short packet period, HSYNC is driven high when a long packet is received.

When the long packet reception is complete, HSYNC is driven low.

Frame start short packets in the receive channel must be received in advance.

Long packets cannot be received from frame end short packets to frame start short packets period.

68.3.14.3 Field Generation in Interlacing

To treat CSI-2 input as interlaced, a field signal is generated using the frame numbers in the frame start and frame end packets.

When a frame start short packet is received, the field is driven high if the field value of the short packet data meets the conditions specified in the field detection control register (CSIFLD).

If the condition is not met, the field is driven low.

Figure 68.20 shows an example of interlaced video.

In this example, the settings are as follows.

CSIFLD .FLD_EN bit is set to 1.

CSIFLD .FLD_SEL bit is set to 01b.

CSIFLD.FLD_NUM bit is set to 0.

Note: Frame numbers are included in FS and FE.

The frame number is incremented by 1 for each FS packet with the same virtual channel and is reset to 1 periodically (e.g., 1, 2, 1, 2, 1, 2).

If the frame number of even field is 0x0001, Set FLD_NUM in CSIFLD to 0x1.

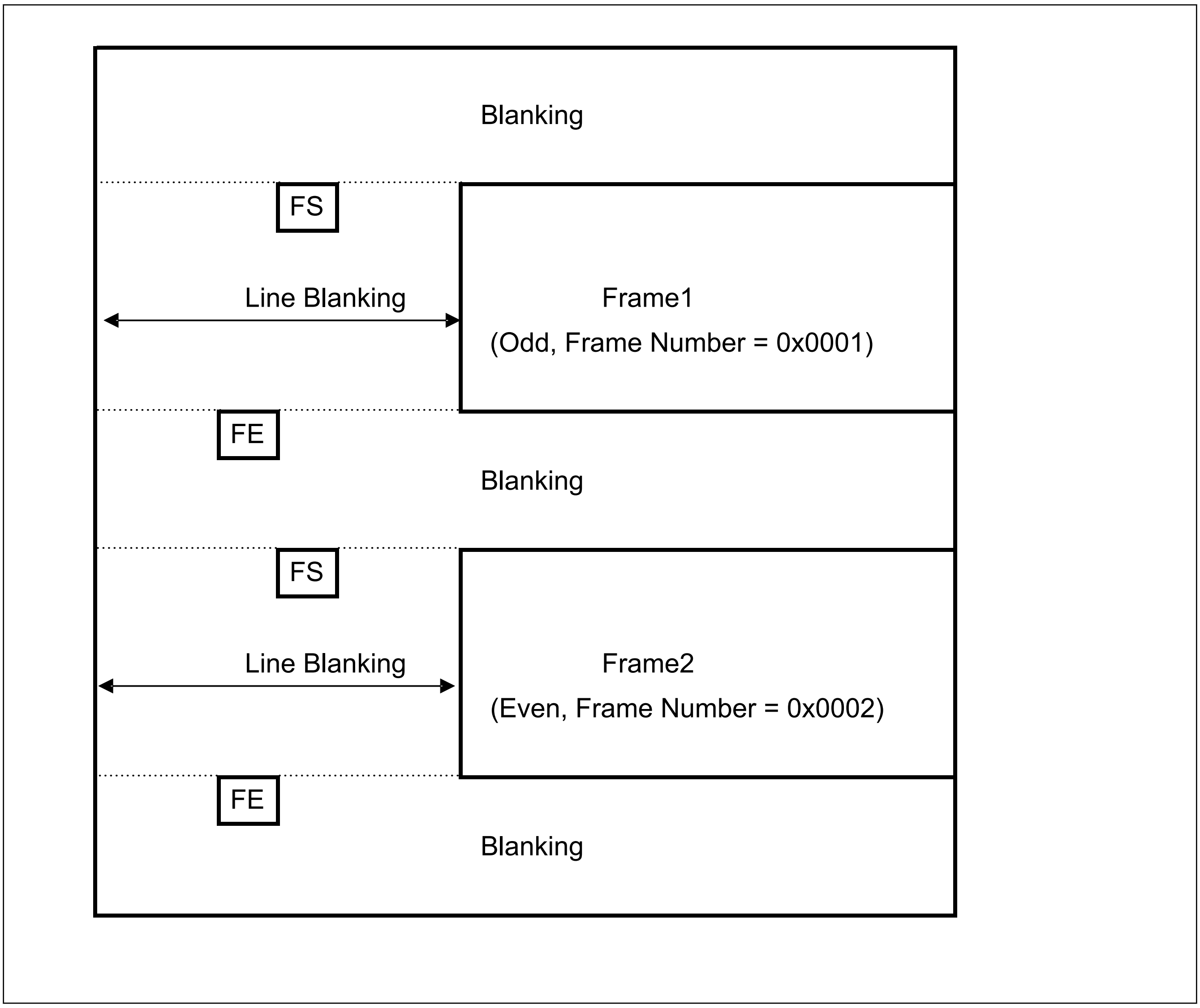


Figure 68.20 Example of interlaced video

68.3.15 Restart Due to Error

Table 68.17 shows the impact of the error on VIN as follows

Table 68.17 Effects of errors (1 of 2)

	Errors	VIN operation	How to deal with it
1	FIFO overflow	Image Disturbance	Restart*1
2	ECC 2-bit error at frame end	Missing image (field)	Not required
3	ECC 2-bit error at frame start	Missing image (field)	Not required

Table 68.17 Effects of errors (2 of 2)

	Errors	VIN operation	How to deal with it
4	ECC 2-bit error for long packets (One line is discarded.)	Image distortion due to missing vertical effective lines	• Restart required if an error is detected in which the effective pixel is shorter than the clip*2. • No action required if an error is detected in which the effective pixel is longer than the clip.
5	Long packet WC error	Image distortion due to missing horizontal valid pixels	
6	AXI Resp Error	No stop or other processing is performed. If data cannot be output to memory, the image will be distorted.	Not required
8	Response over detection	INTS.FMS and MS.MA do not work properly.	If system operation with INTS.FMS and MS.MA is required, restart*3.
10	If some problem occurs in the camera device during the stop process. 1. The video capture active status (CA) bit in the module status register (MS) is not cleared to 0. 2. CSI-2 RX cannot be stopped normally.	Will not stop. Cannot restart.	Figure 68.21 shows the stop process.

Note 1. If a FIFO overflow error occurs, restart the system using the following procedure.
INTS.FOS and interrupt VIN0_IRQ are asserted.
Please take action such as accepting Internal Bus request earlier.

Note 2. If an error occurs in which the clip area is larger than the valid area of the input, restart the system using the following procedure.
INTS.PRCLIPVES and PRCLIPHES interrupts VIN0_IRQ are asserted.

Note 3. If system operation using INTS.FMS and MS.MA is required and a response over error occurs, please follow the procedure below to restart the system. VIN0_IRQ interrupt is asserted when INTS.ROS is set.
Please take measures such as speeding up the response of the Internal Bus.
If INTS.FMS and MS.MA are not used, there is no need to restart.

Restart procedure due to error

For forced termination, set STOPREQ to 1; when AXI stops issuing new requests and all requests in outstandings are completed, STOPACK will be set to 1. After confirming that STOPACK is set to 1, reset VIN. Then follow the initialization procedure to start up

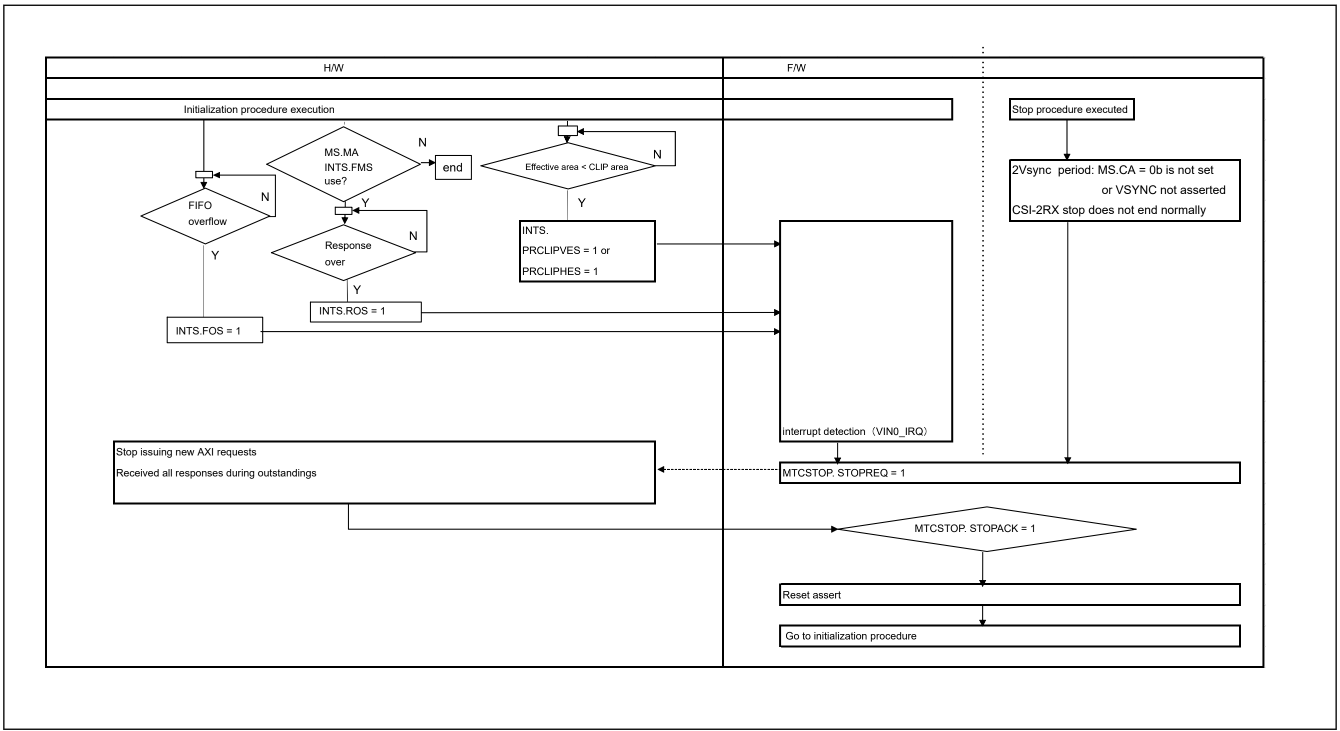


Figure 68.21 Procedure for recovering from an error

68.4 Usage Notes

68.4.1 Specifications

The following shows the specifications.

Table 68.18 Specifications (1/2)

Item	Function
Limitation on register update	If a register is updated during capture, data captured immediately after the register update cannot be guaranteed.
Field capture mode image quality	Images of odd-numbered, odd/even-numbered and even-numbered fields, captured by MC/IM interlace mode bit settings, contains every other line of the input interlaced image. Therefore, note that the horizontal resolution for video display is in units of fields.
Interrupt event timing	INTS.FIS and INTS.EFS are generated by field switching. This interrupt indicates that the transmission is complete from VIN, but does not guarantee that the write to memory is complete. The completion of the write to memory can be checked in INTS.FMS.
Pixel post-clip setting	If the output format is YCbCr422, the clipping size in CL_HSIZE must be set to an even number. Pixel post clipping should be set to the size generated by the scaling parameter. If the horizontal size generated by YCbCr422 is odd, set the generated size -1 (even value).
Scaling up	When horizontal and vertical scaling up is specified, the amount of memory transfer becomes greater with an increase in traffic. Note the amount of traffic on the entire system when using the scaling-up function because the overall transfer efficiency of the system might degrade due to the increased use of the internal buses.
YC Restrictions on YC separation function	Set the offset register (UVAOF) that stores UV data such that the storage areas of Y and UV data are not the same.
RGB-888 → RGB-565/ARGB-1555 conversion function	In the dithering process by cumulative addition, if the same color is captured as in a blue back image, periodic noise may be generated by the cumulative addition process (carried by addition). In this case, set the DC[1:0] bits in MC to the ordered dithering.
Clipping size setting	The common pre-clipping size for odd and even fields is specified in SLPRC, ELPRC, SPPRC, and EPPRC. The common post-clipping size for odd and even fields is specified in UDS_CLIP_SIZE.

Table 68.19 Specifications (2/2)

<table>
<tr><th>Item</th><th>Function</th></tr>
<tr><td>Horizontal clipping specification</td><td>Horizontal clipping size shall be set as follows.
<table>
<tr><th>Capture data</th><th>Pre-clipping start unit</th><th>Clipping size unit</th></tr>
<tr><td>YCbCr-422 data</td><td>2 pixels</td><td>2 pixels</td></tr>
<tr><td>RGB data</td><td>2 pixels</td><td>2 pixels</td></tr>
<tr><td>RAW data</td><td>4 pixels</td><td>4 pixels</td></tr>
</table></td></tr>
<tr><td>MIPI CSI-2's Horizontal blanking</td><td>MIPI CSI-2 horizontal blanking requires at least 20 aclk.</td></tr>
<tr><td>FE to FS interval</td><td>When scaling is not used: For MIPI CSI-2 input, the interval between frame end short packets and frame start short packets should be 1 line or more.
When scaling is used: For MIPI CSI-2 input, the interval between frame end short packets and frame start short packets should be 3 lines or more.</td></tr>
</table>

68.4.2 Power gating control or Software Standby Mode

Before transitioning to software standby mode, stop operation as described in section 68.3.12. Capture Stop Procedure.

After returning from software standby mode, resume operation as described in section 68.3.13. Capture Restarting Procedure.

68.4.3 Module-Stop Function Setting

VIN operation can be disabled or enabled using Module Stop Control Register C (MSTPCRC). The VIN module is initially stopped after reset. Releasing the module-stop state enables access to the registers.